

Monetary Policy Surprises, Credit Costs, and Economic Activity[†]

By MARK GERTLER AND PETER KARADI*

We provide evidence on the transmission of monetary policy shocks in a setting with both economic and financial variables. We first show that shocks identified using high frequency surprises around policy announcements as external instruments produce responses in output and inflation that are typical in monetary VAR analysis. We also find, however, that the resulting “modest” movements in short rates lead to “large” movements in credit costs, which are due mainly to the reaction of both term premia and credit spreads. Finally, we show that forward guidance is important to the overall strength of policy transmission. (JEL E31, E32, E43, E44, E52, G01)

This paper provides evidence on the nature of the monetary policy transmission mechanism. We focus in particular on how monetary policy actions influence credit costs that in turn affect economic activity. Our goal is to assess the extent to which the response of credit costs to monetary policy is consistent with standard theory and, in doing so, identify any significant discrepancies that the theory should address.

There is of course a voluminous literature on monetary policy transmission.¹ Two main considerations motivate us to revisit this classic topic. First, the conventional models of monetary policy transmission treat financial markets as frictionless. To put it mildly, the recent financial crisis suggests rethinking this premise. As we discuss in Section I, the conventional “frictionless” frameworks have sharp predictions for how credit costs should respond to monetary policy actions. In particular, the response of borrowing rates should depend entirely on the expected path of the central bank’s policy instrument, the short-term interest rate. To a first approximation there should be no response in either term premia or credit spreads. We proceed to examine this hypothesis. The goal here is to determine whether a significant component of the response of credit costs to monetary policy may indeed reflect movements in term premia and credit spreads, consistent with some form of financial market imperfection.

*Gertler: Department of Economics, New York University, 19 W. 4th Street, NY, 10003, and National Bureau of Economic Research (NBER) (e-mail: mark.gertler@nyu.edu); Karadi: DG Research, European Central Bank, Neue Mainzer Strasse 66, Frankfurt am Main, 60323, Germany, and Centre for Economic Policy Research (CEPR) (e-mail: peter.karadi@ecb.int). Prepared for the NBER conference on “Lessons From the Crisis for Monetary Policy,” October 18, 19 in Boston. We are grateful to Claudio Schioppa for his excellent research assistance, to Refet Gürkaynak for sharing his data, to an anonymous referee and to Gianni Amisano, Karel Mertens, Giorgio Primiceri, Eric Swanson for helpful discussions. The views expressed are those of the authors and do not necessarily reflect the views of the European Central Bank (ECB) or the Eurosystem.

[†]Go to <http://dx.doi.org/10.1257/mac.20130329> to visit the article page for additional materials and author disclosure statement(s) or to comment in the online discussion forum.

¹See Boivin et al. (2010) for a recent survey.

The second consideration involves the evolution in the way the Federal Reserve manages its policy instrument. In conventional descriptions of the transmission mechanism, the central bank adjusts the current short-term interest rate. Market participants then form expectations about the future path of the short rate based on the central bank's historical tendencies for rate adjustment. However, as Gürkaynak, Sack, and Swanson (2005) (hereafter GSS) and others have argued, over time the Federal Reserve has increasingly relied on communication—in FedSpeak known as “forward guidance”—to influence market beliefs about the expected path of short-term rates. Indeed, once the short term interest rate reached the zero lower bound during the current crisis in December 2008, forward guidance became the only way the central bank could affect market interest rates without resorting to unconventional credit market interventions. Accordingly, in assessing how monetary policy influence credit costs, it is important to account for the role of forward guidance.

To evaluate the nature of monetary policy transmission, we analyze the joint response of a variety of economic and financial variables to exogenous monetary policy surprises. The policy surprises, further, include shocks to forward guidance. Our specific approach involves combining the traditional “money shock” vector autoregression (VAR) analysis (e.g., Bernanke and Blinder 1992, Christiano, Eichenbaum, and Evans 1996)² with high frequency identification (HFI) of the effects of policy surprises on interest rates (e.g., Kuttner 2001; Gürkaynak, Sack, and Swanson 2005; Hamilton 2008; Campbell et al. 2012). GSS, in particular, use unexpected changes in federal funds rate and Eurodollar futures on Federal Open Market Committee (FOMC) dates to measure policy surprises. Our hybrid approach employs HFI measures of policy surprises as external instruments in a set of VARs to identify the effects of monetary shocks. The VARs we consider include output, inflation and a variety of interest rates.

The particular approach we take is dictated by the need to identify policy surprises that can be considered exogenous with respect to both the economic and financial variables in the VAR. Otherwise it is not possible to properly identify the responses of these variables to policy shifts. The standard identification strategy in VARs analyzing monetary shocks is to impose timing restrictions on both the behavior and the impact of the policy rate, typically taken to be the federal funds rate. A standard set of restrictions is to suppose that within a period the funds rate responds to all the other variables in the VAR but not vice versa. That is, the impact of the funds rate on the other variables occurs with a lag of at least one period. So long as the period length is not too long (e.g., a month to a quarter), these kinds of timing restrictions may be reasonable for the interactions between the funds rate and economic activity variables such as output and inflation. However, they are problematic once additional financial variables are present. The problem is simultaneity: Within a period, policy shifts not only influence financial variables, they may be responding to them as well. Even if the central bank is not directly responding to the financial indicators, it may be responding to underlying correlated variables left out of the VAR.

The HFI approach addresses the simultaneity issue by using daily data. In particular, policy shocks are surprises in fed funds futures that occur on FOMC days.

²See also Rudebusch (1998) for a critique of the methodology and an early recommendation of using futures data to measure monetary policy shocks.

To isolate the impact of news about monetary policy, the surprises in futures rates are usually measured within a tight window (e.g., thirty minutes) of the FOMC decision. The dependent variables in the event studies are typically the same day responses in various interest rates and asset returns. The key identifying assumption is that news about the economy on the FOMC day does not affect the policy choice. Only information available the previous day is relevant. Given this assumption, surprises in fed funds futures on FOMC dates are orthogonal to within period movements in both economic and financial variables. One additional benefit of this approach is that the policy surprise measure can include “shocks” to forward guidance. Following GSS, this is accomplished by incorporating in the instrument set surprises in fed funds futures for contracts that expire at a subsequent date in the future. These surprises in principle reflect revisions in beliefs on FOMC dates about the future path of short-term rates.

There are however some limitations to the HFI approach. While it is possible to measure the instantaneous effect of a policy surprise on market interest rates, due to the event study framework it is difficult to identify how persistent the impact is. For a similar reason, it is not possible to examine the response of economic activity variables such as output and inflation. In contrast, our approaches combine the strengths of the VAR and HFI methodologies. We exploit the HFI approach to identify exogenous policy surprises but then use a full VAR to trace out the dynamic responses of real and financial variables.

In addition to the classic VAR and HFI papers mentioned earlier, several other papers are relevant to our analysis. Hanson and Stein (2012) and Nakamura and Steinsson (2013) use HFI to investigate the impact of monetary policy surprises on the real yield curve. The former emphasize the response of term premia. The latter use the event study to identify parameters in a small-scale New-Keynesian model. Other papers have incorporated HFI measures of monetary policy shocks into VARs (e.g., Bagliano and Favero 1999; Cochrane and Piazzesi 2002; Faust, Swanson, and Wright 2004; and Barakchian and Crowe 2013). We differ by employing an external instruments approach that permits sharp testing of conventional theories of the impact of monetary policy on credit costs.

In Section I we describe the conventional monetary transmission mechanism in detail. We derive several testable implications involving the response of credit costs to monetary policy shocks. In Section II we present our methodology. We present our VAR framework and describe in particular how we make use of interest rate futures surprises as external instruments. Finally, we present evidence on the response of credit costs and economic activity to monetary policy in Section III. We begin with a discussion of our choice of interest rate to serve as a policy indicator and of the instruments set we use to identify exogenous monetary policy surprises. We then show that in a VAR with both financial and economic variables, our external instrument approach produces a more convincing set of responses to a monetary shock than does a standard Cholesky identification scheme. An unanticipated monetary tightening produces a significant drop in output and a modest insignificant drop in the price level. In addition, real credit costs increase for all of the securities we consider. Each of these results is consistent with conventional models of the monetary transmission mechanism.

But we also obtain some results that are inconsistent with the standard model. In particular, monetary policy responses typically produce “modest” movements in

short rates that lead to “large” movements in credit costs. As we show, the large movements in credit costs are mainly due to the reaction of both term premia and credit spreads. The baseline model of the transmission mechanisms abstracts from both these considerations. At the same time, the large movement in credit costs may help unlock the puzzle of why seemingly modest movements in short-term rates appear to have a substantial impact on economic activity. To be sure, we find evidence for the kind of price stickiness present in the conventional models. The response of real rates to monetary policy surprises is approximately equal to the response of nominal rates. However, to account for the overall response of credit costs, it may be necessary to amend the model to account for movements in term premia and credit spreads.

Finally, we show that forward guidance is important to the overall strength of policy transmission. Holding constant the size of the response of the funds rate, the impact of a monetary policy surprise on economic activity depends on the degree of news from forward guidance embedded in the shock.

I. The Conventional Monetary Policy Transmission Mechanism: Some Testable Implications

In this section we describe the conventional monetary transmission mechanism and propose several testable implications. We take as an example of the conventional mechanism the one present in the New-Keynesian models used widely by central banks across the globe.³ Within these frameworks aggregate spending depends on current and expected future short-term real interest rates. Transmission of monetary policy then works as follows: The central bank chooses the short-term nominal interest rate i_t each period, which we express in annualized terms. Due to some form of nominal price and/or wage rigidities, control over the nominal rate gives the central bank control over current and expected future real rates, at least for some horizon. It is this leverage over the time path of short term real interest rates that allows the central bank to influence aggregate spending that in turn translates into movements in output and inflation.

Given the expectations hypothesis of the term structure, a way to summarize the impact of monetary policy actions on the path of short term interest rates is to examine the response of the yield curve. A loglinear approximation of an m period zero-coupon government bond yields

$$(1) \quad i_t^m = E_t \frac{1}{m} \left\{ \sum_{j=0}^{m-1} i_{t+j} \right\} + \phi_t^m,$$

where i_t^m is the annual bond yield and ϕ_t^m is the annualized term premium. To a first order, ϕ_t^m is a constant within a local region of the steady state. It follows that variation in long-term rates reflects variation in the path of current and expected future short rates. In this respect, the transmission of monetary policy actions to credit

³ Examples of the conventional New-Keynesian dynamic stochastic general equilibrium (DSGE) model include Christiano et al. (2005) and Smets and Wouters (2007). Variations that allow for financial market frictions include Bernanke et al. (1999), Gertler and Karadi (2011) and Christiano et al. (2014). Examples of models with the risk-taking channel of monetary policy are Drechsler et al. (2014) and Brunnermeier and Sannikov (2011).

costs operates via the yield curve. This link between short and long rates is present in standard New-Keynesian models and is a feature of all conventional models of monetary policy transmission. Equation (1) also makes clear how forward guidance provides the central bank with some leverage over longer maturity interest rates, to the extent it is able to effectively communicate its intentions about the path of future short rates.

Of course what matters for monetary transmission is the behavior of real interest rates. Let π_t be the annualized percent change in the price level between time t and $t + 1$ and let π_t^m be the annualized percent change in the price level between t and $t + m$. Then to a first approximation the real return of the m period nominal bond, can be expressed as the following function of the expected path of short-term real rates, again with the additive term premium ϕ_t^m :

$$(2) \quad i_t^m - E_t \pi_t^m = E_t \frac{1}{m} \left\{ \sum_{j=0}^{m-1} (i_{t+j} - \pi_{t+j}) \right\} + \phi_t^m$$

with $\pi_t^m = 1/m \sum_{j=0}^{m-1} \pi_{t+j}$, and as before ϕ_t^m is constant within a local region of the steady state. As we have noted, the standard theory of monetary transmission presumes that the central bank's adjustment of nominal rates leads to adjustment in real rates due to temporary nominal rigidities that inhibit offsetting movements in inflation. Of course, in the standard model the central bank's leverage over longer maturity rates depends on the degree of price stickiness.

Up to this point we have analyzed the link between monetary policy actions and government bond yields. As we noted earlier, in the standard models of the transmission mechanism financial markets are frictionless. Thus, for given maturity, the interest rate on a private security equals the corresponding government bond rate, up to a first order. The effects of monetary policy actions on the government bond yield curve translate exactly into effects on private borrowing rates. With financial market imperfections present, however, monetary transmission may involve a "credit channel" effects (e.g., Bernanke and Gertler 1995). In particular, with credit market frictions operative, the private annual borrowing rate on an m period security i_t^{mp} exceeds the rate on a similar maturity government bond, adjusting for risk. Let x_t^m denote the external finance premium, i.e., the spread between the private security and government bond rates. Then up to a first order

$$(3) \quad i_t^{mp} = i_t^m + x_t^m.$$

With a credit channel present, tightening of monetary policy not only raises government bond rates but also the external finance premium, which amplifies the overall effect of the policy action on private borrowing rates. The external finance premium increases because the tightening of monetary policy leads a tightening of financial constraints. Theories of the credit channel differ on the precise way central bank interest rate shifts influence credit constraints. A common prediction, however, is that the credit channel magnifies the impact of the interest rate adjustment on private borrowing rates via the impact on credit spreads.

Overall, our analysis leads to three implications of the standard theory that we can test. First, the response of the annual yield on an m period government bond to a surprise monetary policy action should equal the surprise in the average of the annualized current short rate and the expected future short rates $m - 1$ periods into the future, with no response of the term premium. Second, the response of the yield on an m period private security should similarly equal the surprise in the expected path of the short rate over a similar horizon, though in this case with no change in either the term premium or the credit spread. Finally, a surprise monetary policy action should affect real rates as well as nominal rates across a nontrivial portion of the yield curve.

To test the first hypothesis, rearrange equation (1) to obtain the following expression for the term premium on the government bond

$$(4) \quad \phi_t^m = i_t^m - E_t \frac{1}{m} \left\{ \sum_{j=0}^{m-1} i_{t+j} \right\}.$$

One can then obtain the response of the term premium to a monetary policy surprise by using the identified VAR to compute the response of the long rate i_t^m and the path of short rate i_t . Under the null of the standard theory, the response of the term premium should be zero.

To test the second hypothesis, first define the excess return on the m period bond, χ_t^m , as the difference between the market rate i_t^{mp} and the average of current and expected annualized short rates over the life of the bond, as follows:

$$(5) \quad \chi_t^m = i_t^{mp} - E_t \frac{1}{m} \left\{ \sum_{j=0}^{m-1} i_{t+j} \right\}.$$

Then combine equations (3) and (5) to obtain the following expression for χ_t :

$$(6) \quad \begin{aligned} \chi_t^m &= i_t^m - E_t \frac{1}{m} \left\{ \sum_{j=0}^{m-1} i_{t+j} \right\} + x_t^m \\ &= \phi_t^m + x_t^m. \end{aligned}$$

Equation (6) relates the excess return on the private security to the sum of the term premium and the external finance premium. We obtain the response of χ_t^m to a monetary policy shock by summing the impulse responses of the term premium and the external finance premium, which we measure directly using the spread between the rate on the private security and a similar maturity government bond. We measure the response of the term premium exactly as in the previous case.

Finally, we can easily compute the response of real interest rates to monetary policy shocks. We do so by first computing the impulse response of the relevant nominal interest rate and the log price level. We then make use of equation (2) to calculate the response of real interest rates. The interesting issue is how much of the response of nominal rates to monetary shocks reflects movement in real rates across different maturities.

II. Econometric Framework

Our econometric model is vector autoregression with a mixture of economic and financial variables. To identify monetary surprises we use external instruments. Our use of external instruments in a VAR is a variation of the methodology developed by Stock and Watson (2012) and Mertens and Ravn (2013). We describe the approach below.

Let $\mathbf{Y}_t$ be a vector of economic and financial variables, $\mathbf{A}$ and $\mathbf{C}_j \forall j \geq 1$ conformable coefficient matrices, and $\boldsymbol{\varepsilon}_t$ a vector of structural white noise shocks. Then the general structural form of the VAR we are considering is given by

$$(7) \quad \mathbf{A}\mathbf{Y}_t = \sum_{j=1}^p \mathbf{C}_j \mathbf{Y}_{t-j} + \boldsymbol{\varepsilon}_t.$$

Multiplying each side of the equation by $\mathbf{A}^{-1}$ yields the reduced form representation

$$(8) \quad \mathbf{Y}_t = \sum_{j=1}^p \mathbf{B}_j \mathbf{Y}_{t-j} + \mathbf{u}_t,$$

where $\mathbf{u}_t$ is the reduced form shock, given by the following function of the structural shocks:

$$(9) \quad \mathbf{u}_t = \mathbf{S}\boldsymbol{\varepsilon}_t$$

with $\mathbf{B}_j = \mathbf{A}^{-1}\mathbf{C}_j$; $\mathbf{S} = \mathbf{A}^{-1}$. The variance-covariance matrix of the reduced form model equals $\boldsymbol{\Sigma}$.

$$(10) \quad E[\mathbf{u}_t \mathbf{u}_t'] = E[\mathbf{S}\mathbf{S}'] = \boldsymbol{\Sigma}.$$

Let $Y_t^p \in \mathbf{Y}_t$ be the policy indicator, specifically the variable in the structural representation (7) with exogenous variation due to the associated primitive policy shock ε_t^p . We distinguish between the policy indicator and the policy instrument. The latter is the current period short term interest rate (specifically the federal funds rate). In the standard money shock VAR the policy indicator and policy instrument are one in the same, since the structural policy shock corresponds to an exogenous innovation in the current short rate. However, because we wish to include shocks to forward guidance in the measure of the policy innovation, we instead take as the policy indicator a government bond rate with a maturity somewhat longer than the current period funds rate. The advantage of the government bond rate is that its innovations incorporate not only the effects of surprises in the current funds rate but also shifts in expectations about the future path of the funds rate, i.e., shocks to forward guidance. Later in this section we describe formally our distinction between the policy indicator and the policy instrument.

Next, let $\mathbf{s}$ denote the column in matrix $\mathbf{S}$ corresponding to impact on each element of the vector of reduced form residuals $\mathbf{u}_t$ of the structural policy shock ε_t^p .

Accordingly, to compute the impulse responses to a monetary shock, we need to estimate

$$(11) \quad \mathbf{Y}_t = \sum_{j=1}^p \mathbf{B}_j \mathbf{Y}_{t-j} + \mathbf{s} \varepsilon_t^p.$$

Because we are not interested in computing a variance decomposition or the impulse responses to other shocks, we do not have to identify all the coefficients of $\mathbf{S}$, but rather only the elements of the column $\mathbf{s}$.

One can simply use least squares estimation of the reduced form VAR to obtain estimates of the coefficients in each matrix $\mathbf{B}_j$. Some restrictions are necessary, however, to identify the coefficients in $\mathbf{s}$. The standard timing restriction we described earlier amounts to assuming that all the elements of $\mathbf{s}$ are zero except the one that corresponds to the policy indicator. This generates extra restrictions, that are sufficient to identify $\mathbf{s}$.

As we noted earlier, however, this kind of timing restriction is problematic when financial variables appear in the VAR along with the policy indicator. A restriction that an innovation in the policy indicator has no contemporaneous impact on other financial variables is generally implausible. In addition, it is difficult to argue that current policy does not respond to the news contained in financial variables. Accordingly, because we are interested in examining the joint response of economic and financial variables, a different approach to identifying monetary policy surprises is needed. It is for this reason that we instead make use of **external instruments as an identification strategy**.

We begin with a general explanation of the external instruments methodology, before turning to the precise approach we take. Following Stock and Watson (2012) and Mertens and Ravn (2013), let $\mathbf{Z}_t$ be a vector of instrumental variables and let ε_t^q be a vector structural shock other than the policy shock. To be a valid set instruments for the policy shock, $\mathbf{Z}_t$ must be correlated with ε_t^p but orthogonal to ε_t^q , as follows:

$$(12) \quad \begin{aligned} E[\mathbf{Z}_t \varepsilon_t^{p'}] &= \Phi \\ E[\mathbf{Z}_t \varepsilon_t^{q'}] &= \mathbf{0}. \end{aligned}$$

To obtain estimates of the elements in the vector $\mathbf{s}$ in equation (11), proceed as follows: First, obtain estimates of the vector of reduced form residuals $\mathbf{u}_t$ from the ordinary least squares regression of the reduced form VAR. Then let u_t^p be the reduced form residual from the equation for the policy indicator and let $\mathbf{u}_t^q$ be the reduced form residual from the equation for variables $q \neq p$. Also, let $s^q \in \mathbf{s}$ be the response of u_t^q to a unit increase in the policy shock ε_t^p . Then we can obtain an estimate of the ratio $\mathbf{s}^q / s^p$ from the two stage least squares regression of $\mathbf{u}_t^q$ on u_t^p , using the instrument set $\mathbf{Z}_t$.

Intuitively, the first stage isolates the variation in the reduced form residual for the policy indicator that is due to the structural policy shock. It does so by regressing u_t^p

on $\mathbf{Z}_t$ to form the fitted value $\widehat{u}_t^p$. Given that the variation in $\widehat{u}_t^p$ is due only to ε_t^p the second stage regression of $\mathbf{u}_t^q$ on $\widehat{u}_t^p$ then yields a consistent estimate of $\mathbf{s}^q/s^p$

$$(13) \quad \mathbf{u}_t^q = \frac{\mathbf{s}^q}{s^p} \widehat{u}_t^p + \boldsymbol{\xi}_t,$$

where $\widehat{u}_t^p$ is orthogonal to the error term $\boldsymbol{\xi}_t$, given the assumption of equation (12) that $\mathbf{Z}_t$ is orthogonal to all the structural shocks other than the shock to the policy indicator ε_t^p . An estimate for s^p is then derived from the estimated reduced form variance-covariance matrix using equations (10) and (13). We are then able to identify $\mathbf{s}^q$.⁴ Given estimates of s^p , $\mathbf{s}^q$ and $\mathbf{B}_j$, we can use equation (11) to compute responses to monetary policy surprises.

As we discussed, following the HFI literature, the set of potential external instruments we use to identify monetary policy shocks consists of surprises in fed funds and Eurodollar futures on FOMC dates. In particular, let f_{t+j} be the settlement price on the FOMC day in month t for interest rate futures (either fed funds or Eurodollars) expiring in $t+j$; and let f_{t+j-1} be the corresponding settlement price for the day prior to FOMC meeting. In addition, let $(E_t i_{t+j})^u$ be the unexpected movement in the target funds rate anticipated for month $t+j$, with $(E_t i_t)^u = i_t^u$ the surprise in the current short rate. Accordingly, we can express $(E_t i_{t+j})^u$ as the surprise in the futures rate, as follows.⁵

⁴Consider partitioning the vector of reduced form residuals as $\mathbf{u}_t = [u_t^p \mathbf{u}_t^q]'$ and the corresponding matrix of structural coefficients as

$$(14) \quad \mathbf{S} = [\mathbf{s} \quad \mathbf{s}_q] = [\mathbf{S}_1 \quad \mathbf{S}_2] = \begin{bmatrix} s_{11} & \mathbf{s}_{12} \\ \mathbf{s}_{21} & \mathbf{s}_{22} \end{bmatrix},$$

and the reduced form variance-covariance matrix as

$$(15) \quad \boldsymbol{\Sigma} = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}.$$

s^p is identified up to a sign convention and can be obtained by the following closed form solution

$$(16) \quad (s^p)^2 = s_{11}^2 = \Sigma_{11} - \mathbf{s}_{12} \mathbf{s}_{12}',$$

where

$$(17) \quad \mathbf{s}_{12} \mathbf{s}_{12}' = \left(\Sigma_{21} - \frac{\mathbf{s}_{21}}{s_{11}} \Sigma_{11} \right)' \mathbf{Q}^{-1} \left(\Sigma_{21} - \frac{\mathbf{s}_{21}}{s_{11}} \Sigma_{11} \right),$$

with

$$(18) \quad \mathbf{Q} = \frac{\mathbf{s}_{21}}{s_{11}} \Sigma_{11} \frac{\mathbf{s}_{21}'}{s_{11}} - \left(\Sigma_{21} \frac{\mathbf{s}_{21}'}{s_{11}} + \frac{\mathbf{s}_{21}}{s_{11}} \Sigma_{21}' \right) + \Sigma_{22}.$$

The derivation is the straightforward application of the restrictions in 10 noticing that $\left(\Sigma_{21} - \frac{\mathbf{s}_{21}}{s_{11}} \Sigma_{11} \right)' \left(\Sigma_{21} - \frac{\mathbf{s}_{21}}{s_{11}} \Sigma_{11} \right) = \mathbf{s}_{12} \mathbf{Q} \mathbf{s}_{12}'$.

⁵Following Kuttner (2001) and others, we measure the surprise in the target rate using the change in the futures rate as opposed to the difference between the realized target and the futures rate forecast. The reason is to cleanse risk premia in futures from the measure of the unanticipated movement in the target. Assuming that the risk premium for the futures rate does not change in the 24 hours leading up to the FOMC decision, differencing the future rates eliminates the risk premium for the measure of the unanticipated change in the target. See also Piazzesi and Swanson (2008) and Hamilton (2009).

$$(19) \quad (E_t i_{t+j})^u = f_{t+j} - f_{t+j-1}$$

For $j = 0$, the surprise in futures rates measures the shock to the current fed funds futures, which is the case studied by Kuttner (2001).⁶ For $j \geq 1$, the surprise in the expected target rate may be thought of as measuring a shock to forward guidance, following Gürkaynak, Sack, and Swanson (2005). Finally, also following GSS, to ensure that the surprises in futures rates reflect only news about the FOMC decision, we measure these shocks within a 30 minute window of the announcement.

We next discuss exactly how we use interest rate futures as external instruments to identify exogenous monetary policy shocks. Within the baseline set of VARs we consider, we take the one-year government bond rate as the relevant monetary policy indicator, rather than the federal funds rate as is common in the literature. As we suggested earlier, using a safe interest rate with a longer maturity than the funds rate allows us to consider shocks to forward guidance in the overall measure of policy shocks. Under this scenario, a component of the reduced form VAR residual for the one government bond rate is a monetary policy shock that includes exogenous surprises not only in the current funds rate but also exogenous surprises in the forward guidance about the path of future rates.

Our conceptually preferred indicator is the two-year government bond rate based on arguments by Swanson and Williams (2014) and Hanson and Stein (2012) and others who argue the Federal Reserve's forward guidance strategy operates with a roughly two-year horizon. That is, the central bank's focus is on managing expectations of the path of the short rate roughly two years into the future. Bernanke, Reinhart, and Sack (2004) and GSS provide some evidence in support of this view. They find that FOMC statements interpretable as providing forward guidance have a significant impact on futures rates that are relevant to pricing the two-year government bond rate.

We find, however, that interest rate futures surprises that have significant explanatory power for movements in the two-year government bond rate on FOMC dates, are not strong instruments for monthly (reduced form) VAR innovations. By contrast, for the one-year government bond rate, it is possible to find good instruments among the set of futures rate surprises. Accordingly, we use the one-year rate as the policy indicator in our baseline analysis, but then show all our results are robust to using the two-year rate. In the next section we describe in detail the issues involved in the choice of the policy indicator as well as the instrument set.

In the mean time, we can be precise about how the innovation in the one-year government bond incorporates policy surprises that allow for shocks to forward guidance. In particular, given the monthly frequency and our earlier notation (see equation (1)) we can approximate the return on the one-year government bond rate, i_t^{12} as a function of current and expected short rates along with a term premium ϕ_t^{12} , as follows.

⁶To measure the surprise in the futures rate for the current month we need to take account of the timing of when the FOMC date occurs within the month. This is because the futures rate is expressed as an average over all the days of the month. Following Kuttner (2001) we multiply the the surprise in f_t by the factor $\frac{T}{T-t}$, where T is the number of days in the month and t is the number of days elapsed before the FOMC meeting.

$$(20) \quad i_t^{12} = E_t \frac{1}{12} \left\{ \sum_{j=0}^{11} i_{t+j} \right\} + \phi_t^{12}$$

Given equation (20), we can argue that the reduced form VAR residual in the equation for i_t^{12} is equivalent to the month ahead forecast error as follows:

$$(21) \quad i_t^{12} - E_{t-1} i_t^{12} = \frac{1}{12} \sum_{j=0}^{11} \{E_t i_{t+j} - E_{t-1} i_{t+j}\} + \phi_t^{12} - E_{t-1} \phi_t^{12}.$$

The residual for i_t^{12} thus depends on revisions in beliefs about the path of short rates as well as unexpected movements in the current short rate and the current term premium. A monetary policy shock within our framework, accordingly, is a linear combination of exogenous shocks to the current and expected future path of future rates. This contrasts with the conventional literature which considers only shocks to the current short rate. By allowing for shocks that cause revisions in beliefs about the future path of short rates we are able to capture shocks to forward guidance.

Of course, innovations in current and expected future interest rates reflect in part news about the economy that in turn induces the central bank to adjust interest rates. The challenge is to identify the component of these innovations that is due to purely exogenous policy shifts. It is for this reason that we consider a set of surprises in fed funds and Eurodollar futures on FOMC days as external instruments. Doing so allows us to isolate the portion of the innovation in the one-year government bond rate that is due entirely to the exogenous policy surprise. Note that by using surprises for futures contracts settled in subsequent months along with the surprise for the current month, we have instruments for innovations in expected future short rates as well as for the current rate.

III. Data, Estimation, and Results

We analyze monthly data on a variety of economic and financial variables over the period 1979:7 to 2012:6. We choose the starting point to coincide with the beginning of Paul Volcker's tenure as Federal Reserve chair. We do not use the pre-Volcker data based on evidence of differences in the monetary policy regime pre- and post-Volcker (e.g., Clarida et al. 2000). We also explore subsample robustness.⁷

The data of course includes the recent crisis, a period where the short-term interest rate reached the zero lower bound. However, until 2011, our baseline policy indicator, the one-year government bond rate, remained positive, indicating some degree of central bank leverage over this instrument. Swanson and Williams (2014) make the case that the zero lower bound was not a constraint on the Federal Reserve's ability to manipulate the two-year rate. This was probably less true for the one-year rate, though the data suggest flexibility at least until the past year. We address the concern over the zero lower bound by showing our results are robust to

⁷In the working paper version we show that our results are robust to reasonable sample splits, including: 1983:1–2012:6, 1983:1–2008:6, 1979:7–2008:6.

- using the two-year rate as the policy indicator; and
- not including the period of the Great Recession.

It's also true that both real and financial variables exhibited greater volatility during the crisis period. We address this issue by showing in the online Appendix that our results are robust to allowing for a simple form of stochastic volatility.

Our potential instrument set consists of futures rates surprises on FOMC dates used by GSS's event study analysis, including the surprises in the current month's fed funds futures (FF1), in the three month ahead monthly fed funds futures (FF4), and in the six month, nine month and year ahead futures on three month Eurodollar deposits (ED2, ED3, ED4). The instruments are available for us from the period 1991:1 through 2012:6, which is shorter than the sample available for the other series. Accordingly, we use the full sample 1979:7 to 2012:6 to estimate the lag coefficients and obtain the reduced form residuals in equation (8). We then use the instrumental variables and reduced form residuals for the corresponding period to identify the contemporaneous impact of monetary policy surprises (i.e., the vector s in equation (11)).

To illustrate the issues of the choices of a policy indicator and associated instruments and of how our external instruments approach works, we start with a simple VAR. This stripped-down VAR includes two economic variables, log industrial production and the log consumer price index, the one-year government bond rate (the policy indicator), and a credit spread, specifically the Gilchrist and Zakrajšek (2012) excess bond premium. We then construct a baseline VAR that includes additional indicators of credit costs. Finally, we consider additional interest rates by adding them one at a time to the baseline.

In the next subsection we present an analysis of our policy indicator and instrument choice. We then use the simple VAR to show how our external instrument identification works, as well as how it compares with a standard Cholesky identification. In the final subsections we present our baseline VAR along with variants. We proceed to use the framework to present our main analysis of the impact of monetary policy surprises on credit costs.

A. Policy Indicator and Instrument Choice

To evaluate our choice of a policy indicator along with instruments for policy shocks, we begin with a high frequency variant of the external instruments approach that we ultimately use in the monthly VAR. In this high frequency variant, we examine the response of various market interest rates to surprises in various policy indicators, using interest rate futures surprises on FOMC dates as instruments. As in the HFI literature, the dependent variables in this exercise are surprises in daily rates. What this exercise permits is an analysis of the implications of different policy indicators and instruments for market interest rates in a setting where all the instruments have good explanatory power. This then sets the stage for an evaluation of indicator and instrument choice for the monthly VAR.

Let i_t^n be the interest rate on an n month government bond that serves as the policy indicator, let ΔR_t be the change in an asset return on an FOMC day, and let $(i_t^n)^u$ be

the same day unanticipated movement in i_t^n . Accordingly, the equation we consider relates ΔR_t to $(i_t^n)^u$ as follows:

$$(22) \quad \Delta R_t = \alpha + \beta(i_t^n)^u + \varepsilon_t.$$

We estimate equation (22) using two stage least squares with various interest rate futures as instruments. Under our identifying assumptions, the instrumental variables estimation isolates variation in i_t^n due to pure monetary policy surprises that is orthogonal to the error term ε_t , which leads to consistent estimates of β . Note that this formulation is slightly different from the convention in the HFI literature, which regresses the change in asset returns directly on the futures rate surprises. However, as we just noted, the exercise corresponds directly to what we do in the VAR analysis. The difference is that in the latter, the dependent variables are monthly VAR residuals.

We consider three policy indicators: the federal funds rate, one-year government bond rate and the two-year government bond rate. We also consider three different instrument rate combinations:

- the surprise in the current federal funds futures rate (FF1);
- the surprise in the three month ahead futures rate (FF4); and
- the full GSS instrument set.

Our choice of FF4 is based on the strong performance of this variable as an external instrument in the VAR analysis, as we illustrate shortly. Finally, the independent variables we consider are the five, ten, and thirty year government bond rates,⁸ the five-by-five forward rate, the Moody's Baa spread and the 30-year conventional mortgage spread.⁹

As we noted earlier, we measure the surprises in the futures rates within a 30 minute window of the FOMC announcement. The response of the various government bond yields, as well as the changes in the monetary policy indicators are measured in a daily window. Because the markets for the Baa and mortgage securities are less liquid than for government bonds we instead measure the response of the returns on these instruments over a subsequent two week period. We estimate the regressions over the available 1991:1–2012:6 sample and we exclude the 2008:7–2009:6 crisis period with excess financial turbulence.

Table 1 presents the results. Each row represents a particular combination of a policy indicator and instrument set. The coefficient in each column represents the impact of a 100 basis point increase in a given policy indicator (due to an exogenous monetary policy shock) on a corresponding asset return. Overall, three results stand out: First, innovations in the one and two-year government bond rates induced by policy surprises have a stronger effect on longer term interest rates and credit spreads than do similarly induced innovations in the federal funds rate. A surprise 100 basis point increase in the funds rate instrumented by FF1 has a significant effect on both

⁸We use the daily series of constant maturity government bond yields derived by Gürkaynak et al. (2007).

⁹The spreads are calculated over the ten-year government bond rates.

TABLE 1—YIELD EFFECTS OF MONETARY POLICY SHOCKS (*Daily*, 1991–2012)

Indicator and instruments	2 year (1)	5 year (2)	10 year (3)	30 year (4)	5 × 5 forw (5)	Baa ⁺ (6)	Mortg. ⁺ (7)
FF, FF1	0.367*** (3.467)	0.233** (2.241)	0.0980 (1.053)	0.00637 (0.103)	−0.0369 (−0.388)	0.139 (1.475)	0.170 (1.445)
1 YR, FF1	0.739*** (8.493)	0.469*** (3.094)	0.197 (1.173)	0.0128 (0.103)	−0.0744 (−0.379)	0.280 (1.544)	0.343 (1.416)
1 YR, FF4	0.880*** (15.81)	0.683*** (8.201)	0.375*** (4.410)	0.145* (1.694)	0.0668 (0.614)	0.333** (2.176)	0.427** (2.239)
2 YR, FF4		0.778*** (11.80)	0.432*** (5.306)	0.169* (1.839)	0.0848 (0.702)	0.355** (1.986)	0.483** (2.141)
2 YR, GSS		0.878*** (18.70)	0.575*** (11.84)	0.234*** (4.139)	0.271*** (3.601)	0.231* (1.844)	0.350** (2.049)

Note: Robust z-statistics in parentheses; QE dates and crisis period are excluded, 188 observations.

⁺Two-week cumulative changes

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

the two and five-year government bond rate: The former increases roughly 37 basis points and the latter 23. Conversely, a similar increase in the one-year government bond rate also instrumented by FF1 has roughly double the effect on the two and five year rate. Simply put, the one-year rate captures more persistent changes in interest rate policy than the funds rate does. As Kuttner (2001) observes, a nontrivial portion of the variation in the funds rate reflects changes in the timing of the rate adjustment, as opposed to a persistent adjustment in the policy rate.

Second, for a given policy indicator, instruments that reflect expectations of interest rate movements further into the future induce a stronger impact of the policy indicator on market interest rates. For example, instrumenting the one-year government bond rate with FF4 instead of FF1 increases its impact on the two-year rate from 74 to 88 basis points and on the five-year rate from 47 to 68 basis points. In addition, with FF4 as the instrument, the one-year rate also has a significant positive effect on the ten-year rate, the Baa spread and the mortgage spread. (Note that the impact on the spread variables is suggestive of a credit channel effect, as discussed in Section I.) Similarly, the two-year government bond rate with the complete set of GSS instruments exerts the strongest impact on market rates. Intuitively, funds rate surprises on contracts settled further in the future capture movements in the policy indicator associated with more persistent changes in policy.

Third, the one-year and two-year policy indicators have similar quantitative effects on market interest rates, particularly if instruments are used which reflect some degree of forward guidance (i.e., FF4 or GSS). While the two year with GSS has the largest effects, the one year with FF4 has effects of similar magnitude. Interestingly, in each case the policy indicator has a significant impact on the Baa and mortgage spreads that is nearly identical in magnitude.

Of course we are ultimately interested in the impact of the policy indicator on real interest rates. Both Hanson and Stein (2012) and Nakamura and Steinsson (2013) have shown using TIPS data that virtually all the responsiveness of nominal rates to policy surprises on FOMC dates reflects variation in real rates, with a negligible

TABLE 2—TIPS AND BREAKEVEN INFLATION EFFECTS OF MONETARY POLICY SHOCKS
(Daily, 1999–2012)

Indicator and instruments	TIPS 2 year (1)	TIPS 5 year (2)	TIPS 10 year (3)	Bkeven 2 year (4)	Bkeven 5 year (5)	Bkeven 10 year (6)
FF, FF1	0.245 (1.348)	0.263** (2.217)	0.149** (2.287)	0.0427 (0.596)	−0.116 (−1.553)	−0.109** (−2.081)
1 YR, FF1	0.800*** (4.141)	0.639*** (7.606)	0.384*** (6.121)	0.282* (1.913)	−0.0932 (−0.620)	−0.125 (−1.165)
1 YR, FF4	0.804*** (5.171)	0.565*** (5.763)	0.315*** (4.136)	0.0990 (0.474)	0.00376 (0.0269)	−0.0738 (−0.815)
2 YR, FF4	0.759*** (5.090)	0.618*** (4.302)	0.344*** (3.592)	0.0935 (0.525)	0.00412 (0.0269)	−0.0808 (−0.743)
2 YR, GSS	0.754*** (7.749)	0.630*** (8.394)	0.462*** (9.350)	0.196** (1.981)	0.189** (2.165)	0.101* (1.818)

Note: Robust z-statistics in parentheses; QE dates and crisis period are excluded, 58 (2 year), 100 observations.

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

response of expected inflation. Table 2 reports the results from a similar exercise using our instrumental variables methodology. The dependent variables are the TIPS two-year, five-year and ten-year real rates and the corresponding breakeven inflation rates.¹⁰ We also consider the same mix of policy indicators and instruments from Table 1. The five- and ten-year rates are available from 1999:1, and the two-year rates from 2004:1. The results confirm that virtually all the impact of the policy surprise is on real rates, with virtually no impact on inflation. In addition, these results confirm that the main insights from Table 1 also hold in this context. The one- and two-year rates as policy indicators have a much stronger impact on market interest than does the funds rate. In addition, the one and two-year rates have a nearly identical effect on real rates and expected inflation. One small qualification is that the two-year rate with the GSS instrument set has a small but significant effect on inflation.

We now turn to the issue of policy indicator and instrument choice in the monthly VARs. While it appears possible to use the one- and two-year rates interchangeably with high frequency dependent variables, a complication emerges in the monthly VARs. In particular, the futures rate surprises on FOMC days appear to be good instruments for the monthly VAR innovation in the one-year rate, but they may be less effective as instruments for the two-year rate.

Table 3 summarizes the issues. The columns considered are the first stage regression residual of a particular policy indicator regressed on various instrument sets.¹¹

¹⁰The rates are from the daily TIPS yield curve estimates of Gürkaynak, Sack, and Wright (2010).

¹¹For the monthly VAR, we need to turn the futures surprises on FOMC days into monthly average surprises. If all the FOMC meetings were on the first day of each month, our job would be easy, the surprises on those days would be our measure of monthly average surprise. But, in reality, the day of the FOMC meetings vary over the month, and we do not want to lose information by disregarding this. Furthermore, as we use monthly average rates (not end of the month rates) for our monetary policy indicators, a surprise that happens at the end of a month can be expected to have a smaller influence on the monthly average rate than a surprise coming at the beginning of the month. We can do our calculation in two steps. First, for each day of the month, we cumulate the surprises on any FOMC days during the last 31 days (e.g., on February 15, we cumulate all the FOMC day surprises since January 15), and, second, we average these monthly surprises across each day of the month. Or, equivalently, we can first create a cumulative daily surprise series by cumulating all FOMC day surprises (similarly as was done by

TABLE 3—EFFECTS OF HIGH-FREQUENCY INSTRUMENTS ON THE FIRST STAGE RESIDUALS OF THE FOUR VARIABLE VAR (*Monthly*, 1991–2012)

Variables	1 year (1)	1 year (2)	1 year (3)	1 year (4)	1 year (5)
FF1	0.890*** (4.044)				0.394 (1.129)
FF4		1.151*** (4.184)		1.266*** (4.224)	1.243*** (3.608)
ED2					1.440 (1.244)
ED3					−4.443*** (−2.635)
ED4			0.624** (2.039)	−0.167 (−0.476)	2.674** (2.493)
Observations	258	258	258	258	258
R^2	0.066	0.078	0.025	0.079	0.110
F -statistic	16.36	17.50	4.159	11.00	8.347
Variables	2 year (6)	2 year (7)	2 year (8)	2 year (9)	2 year (10)
FF1	0.533** (2.116)				0.174 (0.462)
FF4		0.779** (2.272)		1.013*** (2.643)	1.379*** (3.361)
ED2					1.134 (0.859)
ED3					−4.733** (−2.448)
ED4			0.293 (0.923)	−0.339 (−0.863)	2.946** (2.465)
Observations	258	258	258	258	258
R^2	0.020	0.029	0.005	0.033	0.064
F -statistic	4.477	5.160	0.851	3.760	5.162

Note: Robust t -statistics in parentheses.

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

The residuals are computed from the simple VAR described earlier that includes industrial production, the consumer price index, the excess bond premium and a policy indicator.¹² The first five columns consider the one-year rate as the policy indicator, while the last five consider the two-year rate. Both the R^2 and the robust F -statistic for each regression are reported at the bottom of the corresponding column.

To be confident that a weak instrument problem is not present, Stock et al. (2002) recommend a threshold value of ten for the F -statistic from the first-stage regression. Table 3 shows that in three of the five cases for the one-year rate, the F -statistic is safely above this threshold. The instrument that works best is FF4, which explains nearly 8 percent of monthly innovation in the one-year rate and has an associated

Romer and Romer 2004 and Barakchian and Crowe 2013), then, second, we can take monthly averages of these series, and, third, obtain monthly average surprises as the first difference of this series.

¹²The results are robust to using the richer VARs described in the next section.

F -statistic of the seventeen and a half. For the two-year rate, none of the instrument combinations meets the threshold. This is somewhat surprising, given the strong explanatory of the GSS instrument set, and ED4 in particular, for the variation in the two-year rate in the high frequency data. One possibility is that as compared to the one-year rate and FF4, there is greater high frequency variation in the two-year rate and ED4 in daily data. Conversely, movements in the one-year rate and FF4 are relatively persistent by comparison.

The evidence in Tables 1, 2, and 3 leads us to choose the one-year rate as the policy indicator and the three month ahead funds rate surprise (FF4) as the policy indicator for our baseline case. This permits us to establish a set of results for our external instrument approach in a setting where there is unlikely to be a weak instruments problem. We then show that our results are robust to variations that allow for the two-year rate as the policy indicator. We also explore variations in the instrument set that capture different degrees of forward guidance.

B. Results from the Simple VAR

To show how our external instruments approach works, we first present results from the simple VAR that includes the log industrial production, the log consumer price index, the one-year government bond rate as the policy indicator, and the GZ excess bond premium. Roughly speaking, the latter is the component of spread between an index of rates of return on corporate securities and a similar maturity government bond rate that is left after the component due to default risk is removed. As such, it is possibly interpretable as a pure measure of the spread between yields on private versus public debt that is due to financial market frictions. The inclusion of the excess bond premium allows us to clearly illustrate that our external instruments approach is particularly useful when examining the response of financial as well economic variables to exogenous monetary policy surprises.

We choose the excess bond premium as the financial indicator in the simple VAR for two reasons. First, as GZ show, the excess bond premium has strong forecasting ability for economic activity, outperforming every other financial indicator. Accordingly, this variable may provide a convenient summary of much of the information from variables left out of the VAR that may be relevant to economic activity. Second, we are ultimately interested in examining the response of credit costs to monetary policy surprises. As we show, it is fairly straightforward to evaluate the response of the excess bond premium against the conventional theory of monetary policy transmission, particularly since the measure cleans off default considerations.

Figure 1 shows the impulse responses of both the economic and financial variables in the simple VAR. The left panels show the case where money shocks are identified using external instruments. For comparison, the right panels show the case using a standard Cholesky identification. In each case, the panels report the estimated impulse responses along with 95 percent confidence bands, computed using bootstrapping methods.¹³

¹³ Similarly to Mertens and Ravn (2013), we are using wild bootstrap that generates valid confidence bands under heteroskedasticity and strong instruments. The estimation errors related to the instrumental variable

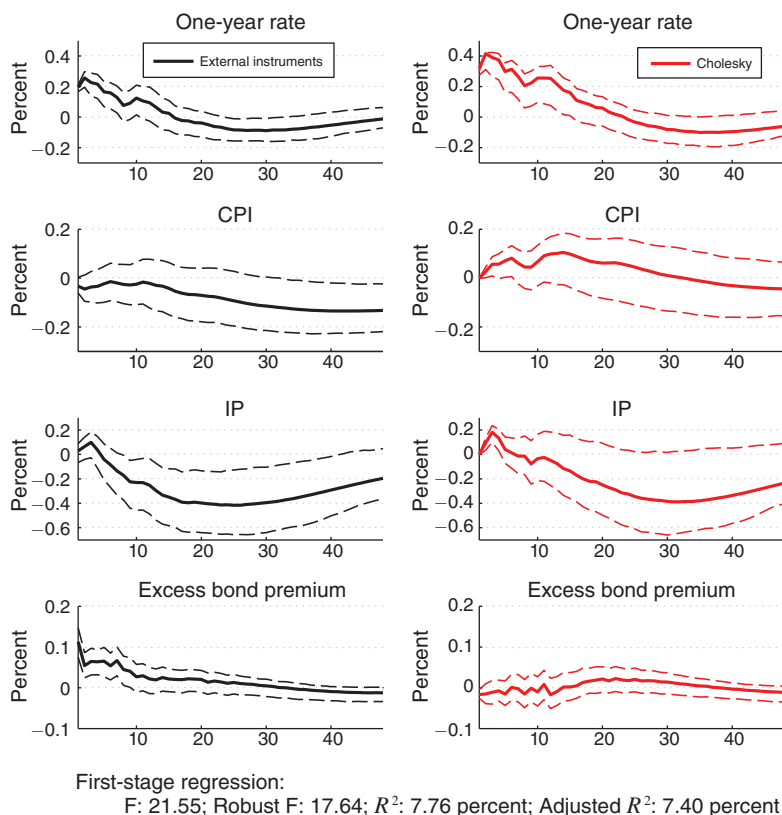


FIGURE 1. ONE-YEAR RATE SHOCK WITH EXCESS BOND PREMIUM

We begin with the external instruments case. As noted earlier, we use the three month ahead funds rate future surprise FF4 to identify monetary policy shock. As a check to ensure that this instrument is valid, we report the F -statistic from the first stage regression of the one-year bond rate residual on FF4. We find an F -value of 21 and half. We also compute a robust F -statistic (which allows for heteroskedasticity) of 17.5. Both values are safely above the threshold suggested by Stock et al. (2002) to rule out a reasonable likelihood of a weak instruments problem.

As the top left panel shows, a one standard deviation surprise monetary tightening induces a roughly 25 basis point increase in the one-year government bond rate. Consistent with conventional theory, there is a significant decline in industrial production that reaches a trough roughly a year and a half after the shock. Similarly consistent with standard theory, there is a small decline in the consumer price index that is not statistically significant. Note that in contrast to the Cholesky identification, we do not impose zero restrictions on the contemporaneous responses of output and inflation. The identification of the monetary policy shock is entirely due to the external instrument.

regression is incorporated in the reported confidence bands, because both stages of the estimation are included in the bootstrapping procedure. Thereby, we avoid any potential "generated regressor" problem.

The GZ excess bond premium increases on impact roughly 10 basis points, an amount which is statistically significant. The spread further remains elevated at roughly 7 basis points for roughly another half year. As we elaborate in the next section, this increase in the excess bond premium following the monetary tightening is consistent with a credit channel effect on borrowing costs: It cannot be explained simply by an increase in bankruptcy probabilities since default premia have been cleaned off the measure.

Critical for obtaining these results is our use of external instruments to identify monetary shocks, as comparison with the case of the Cholesky identification scheme makes clear. Under the Cholesky scheme we consider, the one-year rate is ordered second to last and the excess bond premium last. Given these identifying restrictions, the central bank adjustment of the one-year rate can have an immediate impact on the excess bond premium. But contemporaneous innovations in the excess bond premium do not affect how the central bank manipulates the one-year rate. By assumption, any information in the innovation to the excess bond premium that is relevant to the central bank's decision is already contained in the innovations to industrial production and the CPI. Finally, as in the standard Cholesky case, the central bank can respond to news about output and prices within the period but can affect these variables only with a lag.

As Figure 1 shows, how well a pure monetary policy surprise is identified under the Cholesky scheme is questionable. While the one-year rate increases, both industrial production and the CPI display "puzzles": The monetary policy shock induces a modest but statistically significant increase in each variable. Though the point estimate of industrial productions eventually falls, the decline is not statistically significant.

The behavior of the excess bond premium illustrates the problem with the identification. In response to the surprise monetary tightening, this spread exhibits a statistically significant decline of several basis points. The decline in the spread is inconsistent with theory, which would suggest if anything just the opposite. More likely, there is a problem with the identifying restrictions. It appears the central bank is responding at least in part to the information contained in the excess bond premium. The below trend value of the premium is consistent with a strong economy, which induces the central bank to tighten. The resulting output and price puzzles are consistent with this interpretation. Accordingly, the restriction that the central bank does not respond to innovations in the excess bond premium either directly or indirectly (by responding to variables outside the VAR, which are correlated with the premium) does not appear to be reasonable. Conversely, it is not reasonable to assume that the excess bond premium cannot respond instantly to a monetary shock. Thus, we conclude that the conventional timing restrictions used in monetary VARs do not work well in circumstances like the one here, where both financial and real variables are present. By contrast, our external instruments approach appears well suited for this kind of environment.

C. The Response of Credit Costs to a Monetary Surprise: Facts and Interpretation

We are now able to address the central question of the paper: namely, how credit costs respond to exogenous surprises in monetary policy. We do so by examining the

responses of various credit spreads, various government bond yields, and the federal funds rate. We then use this information to construct the response of overall credit costs for different classes of securities along with a decomposition of the movement of these costs between: variation in

- the expected path of short rates;
- term premia; and
- credit spreads.

We present a sequence of VARs rather than a single one that includes all the variables. Because interest rates of varying maturity are highly correlated, including all at once would lead to a problem of multi-collinearity as well as over-parametrization. Accordingly, we begin with a baseline VAR that includes economic activity variables, a variety of credit spreads, and the one-year government bond rate, which we again take as the policy indicator. We choose this specification as our baseline because it appears reasonably robust to subsample splits and the inclusion of additional variables. We then consider a sequence of VARs which adds an additional interest rate to the baseline specification. As we show in the online Appendix, the behavior of the variables in the baseline is essentially invariant to the inclusion of the additional interest rates in the subsequent VARs. Thus, in effect, the subsequent VARs reveal the behavior of the additional interest rate, holding constant the behavior of the variables in the baseline.

We include six variables in the baseline specification: industrial production; the consumer price index; the one-year government bond rate; the excess bond premium, the mortgage spread, and the commercial paper spread. It is the extension of the model of Figure 1 with the addition of the latter two spreads. The three spread variables reflect (components of) credit costs for three significant financial markets. The excess bond premium is relevant to the cost of long term credit in the non-farm business sector. The mortgage spread is relevant to the cost of housing finance. Finally, the three-month commercial paper spread is relevant to the cost of short term business credit, as well as the cost of financing consumer durables.

As earlier, we start with FF4 as the instrument to identify monetary policy surprises. As was the case with the simple VAR, the *F*-statistics from the first stage regressions are safely in the region where a weak instruments problem is very unlikely. We also explore other policy indicator and instrument combinations.

Figure 2 shows the responses of the variables in the baseline to a one standard deviation contractionary monetary policy surprise. The one-year rate increases roughly 20 basis points on impact and then reverts back to trend after roughly a year. There is a significant and fairly rapid drop in industrial output that begins after several months and reaches a trough after roughly 18. The CPI declines steadily, though this decline is not significant. Associated with the output decline is a significant increase, both statistically and quantitatively, in each of the credit spreads. The excess bond premium increases 8 basis points on impact and remains at that level for roughly eight months before returning to trend. The mortgage spread increases only 2 to 3 points on impact but then increases sharply to 7 points above trend after two months. Finally, the commercial paper spread increases roughly 5 basis points on

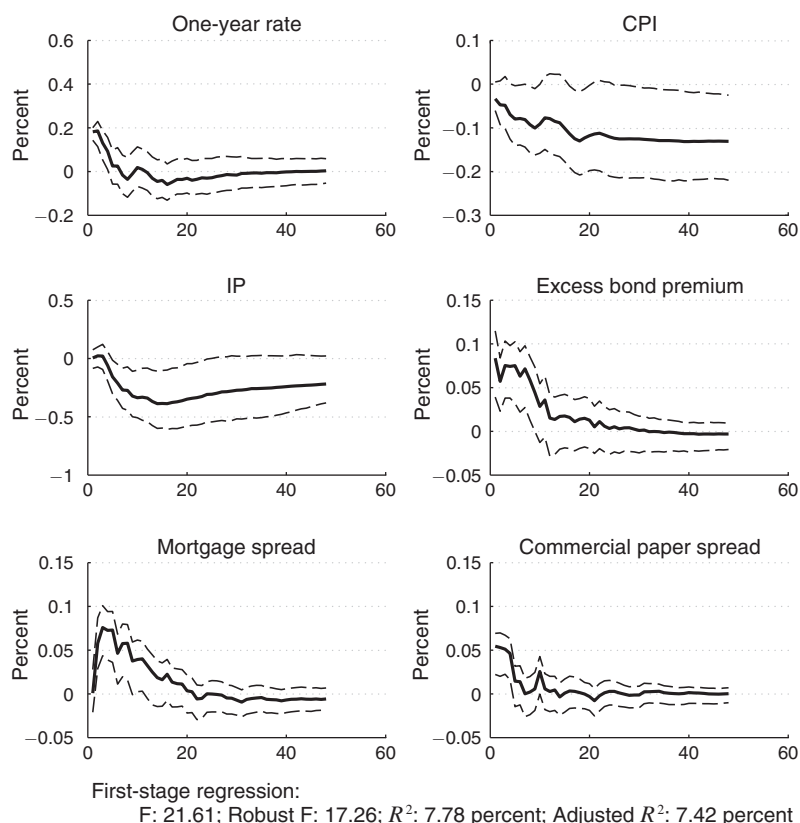


FIGURE 2. ONE-YEAR RATE SHOCK WITH CORPORATE AND MORTGAGE PREMIA

impact for roughly four to five months. We discuss shortly the implications for the movements in these spreads for overall credit costs.

Figure 3 reports the responses of various market interest rates. As we discussed earlier, the responses are calculated by adding each interest rate to the baseline VAR, one at a time. For convenience, we also report the response of the one-year government bond rate from the baseline VAR. The federal funds rate increases significantly upon impact. The increase of roughly 18 points matches the rise in the two-year rate. Both interest rates revert to trend after roughly eight months. This decline is consistent with the quick and sharp contraction in industrial production, presuming the central bank typically reduces short-term rates as the economy weakens. The response of the longer maturity government bond rates is largely consistent with the HFI analysis presented in Table 1. The two-, five-, and ten-year rates increase significantly upon impact but by smaller amounts as the maturity lengthens. Interestingly, all the interest rates revert to trend after about eight months. Finally, also consistent with the HFI analysis in Table 1, the five-by-five forward does not respond significantly upon impact to the surprise tightening. This suggests that the movement in market reflects some combination of expected movements in short rates less than five years out and/or movement in term premia. We turn to this issue in a moment.

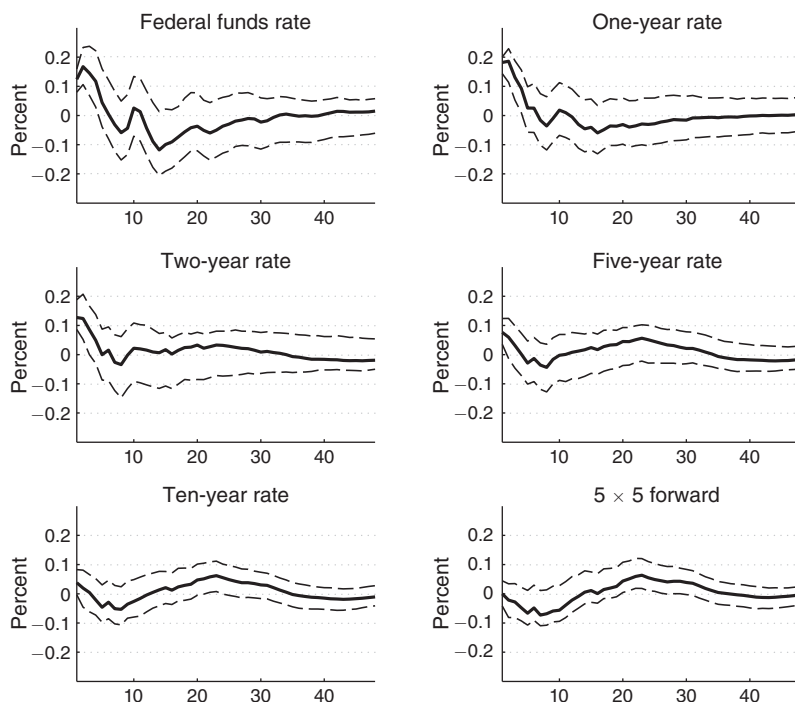


FIGURE 3. ONE-YEAR RATE SHOCK: RESPONSE OF OTHER INTEREST RATES

As we discussed earlier, what matters for monetary policy transmission is the behavior of real rates. Further, HFI analysis using TIPS data suggests that virtually all the movement in nominal rates following a monetary policy surprise reflects movements in real rates, with relatively little impact on inflation. Figure 4 shows that our VAR analysis confirms this result. The left panels report the responses of both the real and nominal two-, five-, and ten-year government bonds rates to a surprise monetary tightening. The right panels report the corresponding breakeven inflation rates. The real interest rate is computed using the response of the nominal rate and expected inflation, the latter calculated from the impulse response of the price level.¹⁴ As the figure shows, real rates move with nominal rates virtually one-for-one. There is a tiny decline in breakeven inflation rates at the different horizons. The declines are marginally significant upon impact at the five- and ten-year horizons, but otherwise insignificant.

We now turn to the question of how overall credit costs respond to monetary policy and what are the factors that underlie the movement in these costs. To address this issue, Figure 5 presents a decomposition of the response of interest rates to the tight money shock. Each of the top four panels reports the response of a government bond rate along with the response of its respective term premium. We calculate the

¹⁴ In particular, we use the price level responses in each corresponding seven variable VARs including the particular nominal rate (not shown). Extending the number of variables introduces minor variations in the estimated price level responses relative to the baseline 6 variable VAR presented on Figure 2, but it does not influence our qualitative conclusions.

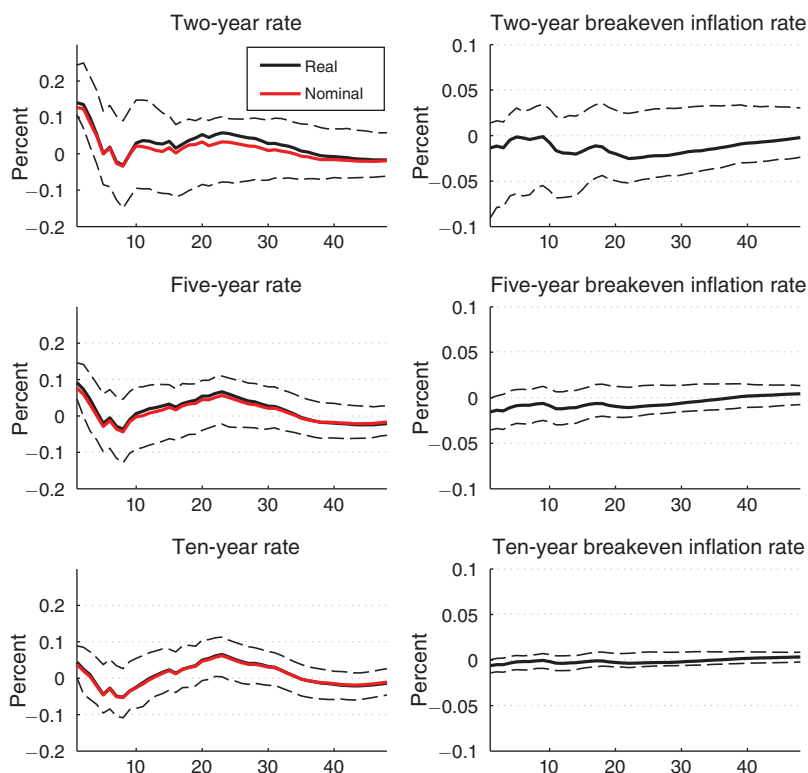


FIGURE 4. ONE-YEAR RATE SHOCK: RESPONSE OF REAL RATES AND BREAKEVEN INFLATION RATES

latter using equation (4), which equates the nominal rate to the sum of the term premium and the expected path of short rates. We use the impulse response of the short rate to compute the latter.

As the figure shows, for the two-year, five-year, and ten-year maturities, virtually all the rate increase is due to the term premium, with no impact of the path of expected short rates. Even for the one-year rate, most of the movement, roughly 80 percent, is due to a term premium effect. That the strong term premium effects arise is not surprising given the behavior of short rates. As we noted earlier, short rates quickly revert to trend following a tight money shock due to the weakening of the economy. It is true that the term premium effects dissipate quickly and are not statistically significant for the ten-year government bond rate. Nonetheless, they are significant on impact for the five-year rate and at least three to four months for the one- and two-year rates.

The bottom two panels of Figure 5 report a similar decomposition of the response of private credit costs, except allowing for a component due to the movement in credit spreads. In particular, each panel reports the response of the combined excess premium for each security along with the overall response of the rate. It is first worth noting that the responses of the corporate bond rates and mortgages are relatively substantial. A roughly 20 basis point increase in the one-year government bond rate leads to a roughly 15 basis point increase in the corporate bond rate and 7 basis point

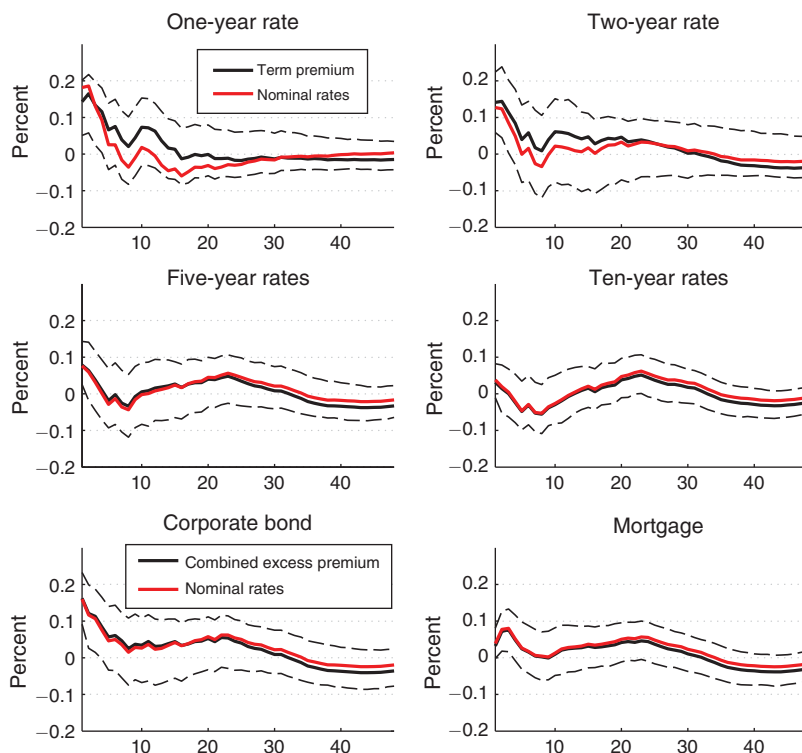


FIGURE 5. ONE-YEAR RATE SHOCK: RESPONSE OF TERM PREMIA AND EXCESS PREMIA

increase in the mortgage rate. In each case virtually all the movement in rates is due to the excess premium, defined as the sum of the credit spread and the term premium. Again, the relatively transitory response of short rates account for the negligible impact of expected short rates on credit costs.

It is of course possible that a component of the term premium response is due to the VAR forecast of short rates not accurately mirroring true market expectations. It could be the case that following a contractionary monetary policy shock, market expectations of the path of interest rates exceed our VAR model's implied path. One possible reason for this overreaction could be that private individuals fail to anticipate the decline in short rates that accompanies the subsequent recession. In this instance, the overreaction of expectations could explain the rise in interest rates across the yield curve, with little or no movement in term premia.

We explore this issue including a survey measure of private sector expectations into our baseline VAR.¹⁵ In particular, we use the Blue Chip Economic Indicators survey, which collects expectations of key market analysts every month since 1983:3 on various interest rates over the subsequent five to six quarters. To facilitate comparison with our baseline one-year policy indicator, we create a one-year constant maturity expectation from the projected path of the federal funds rates. This rate

¹⁵ Nakamura and Steinsson (2013) similarly use survey data to explore term premium effects of monetary policy shocks.

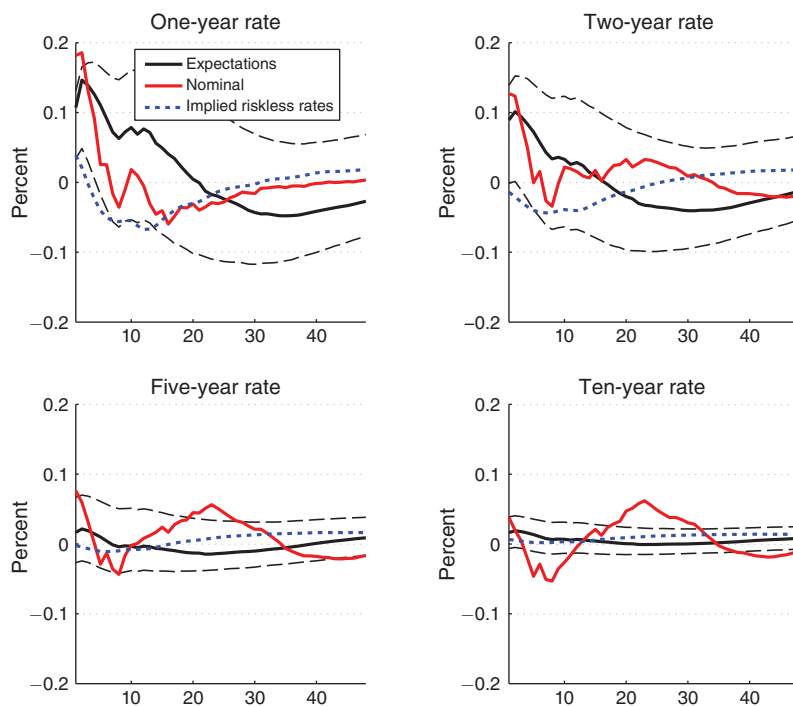


FIGURE 6. ONE-YEAR RATE SHOCK: RESPONSE OF PRIVATE SECTOR EXPECTATIONS, 1979–2012

measures the average short-term interest rate over the subsequent year expected by the private sector.¹⁶ The survey does not collect interest rate expectations beyond six quarters, so to get estimates on the expected path further out, we use our VAR model. In particular, we generate estimates for future expectations by obtaining the VAR implied path of our yearly expectations measure. For average expectations over the two-year horizon, for example, we take the average of the current one-year expectation and its VAR implied level one year ahead. For the average five- and ten-year expectations, analogously, we take the current and the one to four and one to nine year ahead implied rates, respectively.

Figure 6 shows the estimated effects of the monetary policy shock on the average short-term rate expectations over one-, two-, five-, and ten-year horizons. We compare these expectations to similar averages obtained from the “riskless” federal funds rate path of our baseline VAR and to the similar maturity Treasury Bond rates. For the one- and two-year horizons, there is some evidence that market expectations overreact. Their responses exceed the corresponding VAR implied average policy rate, which lies slightly below confidence bands around the path of expectation. Though they also fall short of the corresponding Treasury rate responses, implying, even after correcting for expectations, a small term premium effect. On the other

¹⁶To obtain a yearly measure, we need implied expectations for the last months in the current quarter and the first months in the 4. quarter ahead. For the current quarter, we are using realized rates to filter out expectations for the remaining months and assume constant monthly expectations during the 4. quarter ahead.

hand, the uncertainty surrounding the estimates of the path of expectations makes it very hard to reach definitive conclusions about whether some kind of market overreaction of beliefs (relative to the VAR prediction) could account for the response of term premia at the one- and two-year horizons. At the five- and ten-year horizons, however, market expectations measures line up closely with the implied VAR measures. This suggests that the term premia responses of the five- and ten-year rates are unlikely to be explained by an overreaction of private sector beliefs.

It is important to add that even absent a significant term premium response, our results suggest that the impact of monetary policy on credit costs is stronger than a frictionless model would predict due to sizable effects in credit spreads. We note also that borrowers cannot escape these effects on excess premia for longer term credit by switching to short term since the spread for short-term securities (the commercial spread) increases as well, as Figure 4 shows.

We next turn to the issue of the role of forward guidance. A conceptually nice way to assess the importance of forward guidance would be to follow GSS by isolating the component of the instrument set that reflects surprises in future rates that are orthogonal to surprises in the current rate. This component, which GSS refer to as the “path” factor then in principle captures the effect of pure shocks to forward guidance. The residual component which GSS refer to as the “target factor” captures shocks associated with movement in the current target rate. Unfortunately, this decomposition between target and path factors leads to instruments that are too weak in the context of our external instruments setup to credibly identify pure surprises to forward guidance.

Accordingly, to get a sense of the importance of forward guidance, we opt for the following indirect approach: We replace the one-year government bond rate with the federal funds rate in our baseline VAR. We also replace the three month ahead futures rate surprises with the contemporaneous surprise in fed fund futures. Roughly speaking, the funds rate instrumented by FF1 is likely to capture less persistent variation in short-term rates than is the one-year rate instrumented by FF4. The results from the HFI analysis in Table 1 are consistent with this interpretation: the one-year rate instrumented by FF4 has a stronger effect on longer term interest rates than does the funds rate instrumented by FF1. This suggests the latter may incorporate a greater degree of forward guidance than the former.

Figure 7 reports the response of the system to a tight money shock in the case with the funds rate as the policy indicator. For comparison, we also report the response of our baseline VAR with the one-year rate as the policy indicator. We normalize the size of the shock so that the response of the funds rate is quantitatively the same as in the baseline case. Overall, the pure funds rate shock has weaker impact on both economic and financial variables than in the case where the innovation in the one-year rate (instrumented by FF4) incorporates the policy surprise. The contraction in output is more than 50 percent smaller and the fall in the CPI is similarly smaller relative to the baseline. In each case, further, the response of the variable in the baseline lies below the 95 percent confidence interval for the funds rate shock. In addition the response of credit spreads and long term interest rates is significantly weaker in response to the funds rate shock relative to the baseline one-year rate shock. Overall, the results are consistent with the impact of the monetary policy surprise increasing

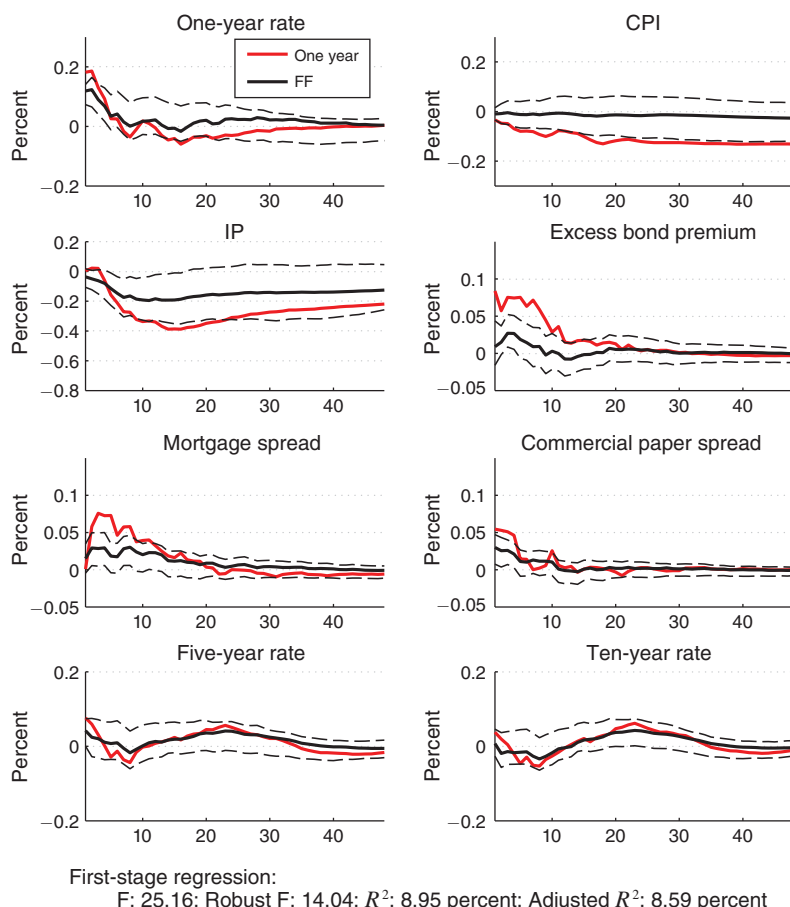


FIGURE 7. FORWARD GUIDANCE: FEDERAL FUNDS VERSUS ONE-YEAR RATE SHOCK

in the relative degree of forward guidance that underlies the shock (holding constant the change in the current funds rate).¹⁷

We conclude with three robustness exercises. First, we replace the one-year rate as the policy indicator, with the two-year rate instrumented by the full GSS set of futures rate surprises. As we noted earlier, the latter is our conceptually preferred policy indicator/instrument variables combination. While concern over weak instruments kept us from using this combination in the baseline VAR, it is useful to assess its performance relative to the baseline. Figure 8 reports the impulse responses of the main variables in our baseline framework for this case. We normalize the two-year

¹⁷Our results are robust to restricting the identification sample to the precrisis period (1991:1-2008:6). This dissolves potential concerns that by using the three month futures surprise (FF4) as instrument, we implicitly overweigh the postcrisis sample, when these rates can be expected to exhibit excess volatility over the current month futures surprises (FF1) as a result of more prevalent forward guidance shocks. In particular, high frequency responses reported in Table 1 are robust to excluding the postcrisis period from the sample. Furthermore, qualitative conclusions drawn from Figure 7 remain intact if we restrict the sample used for identification to the precrisis period. Results are available from the authors upon request.

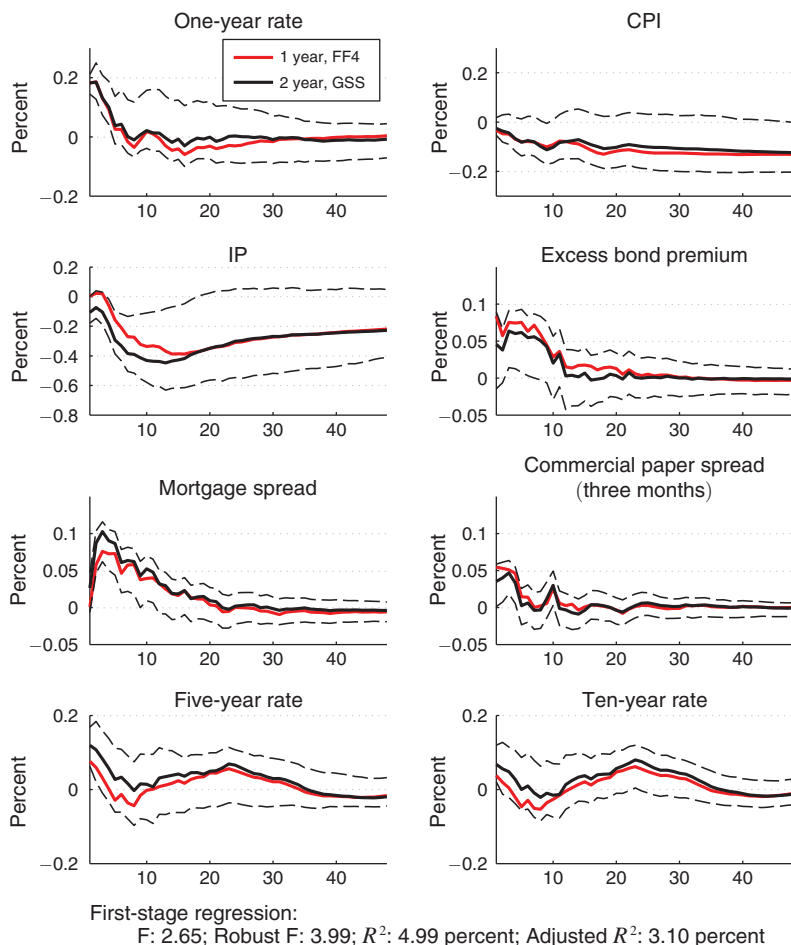


FIGURE 8. TWO-YEAR RATE SHOCK WITH A FULL SET OF GSS INSTRUMENTS

rate shock so that it produces the same movement in the one-year rate as does the shock in the baseline case. Overall, despite the potential weak instruments problem, the two-year rate shock produces responses of the economic variables that are very similar to the baseline. If anything, the two-year rate shock has a slightly stronger effect, though the differences are not significantly different. The stronger effects are consistent with the two-year rate shock having a greater degree of forward guidance, as confirmed by the HFI analysis in Table 2.

Next, we show that our results are not dependent on including the crisis period in the sample. Figure 9 repeats the baseline exercise with a sample that is truncated at 2008:6. As the figure shows, a surprise monetary tightening produces responses of the economic and financial variables that are very similar to the baseline. The drop in output is a bit smaller relative to the baseline, though it remains statistically significant.¹⁸

¹⁸Including a zero restriction on the contemporaneous output response enhances the decline.

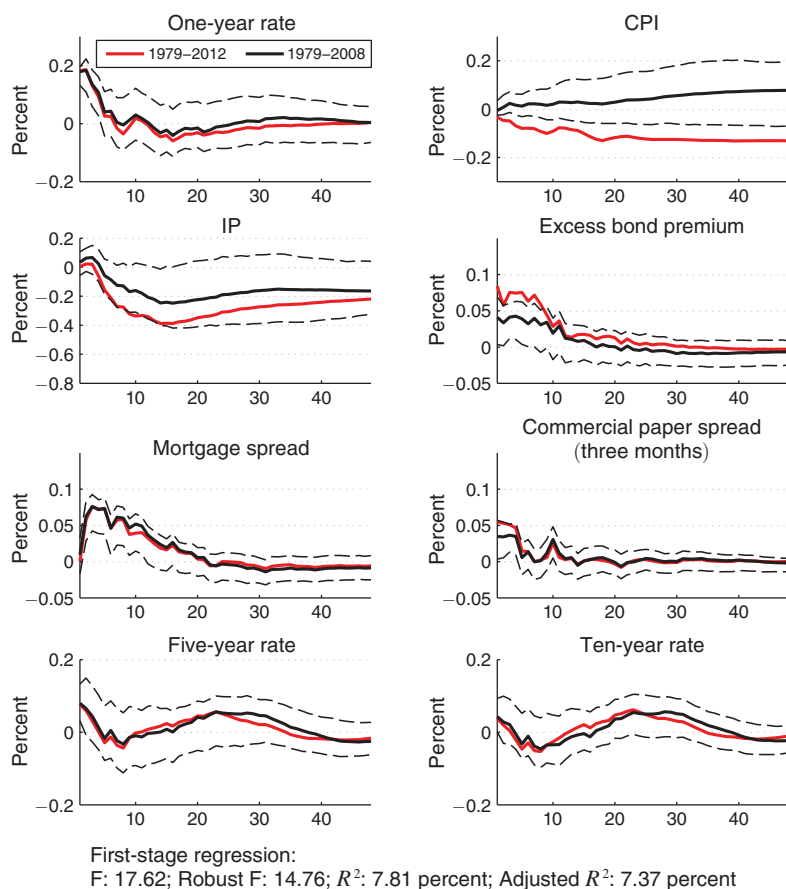


FIGURE 9. ONE-YEAR RATE SHOCK, 1979–2008

Finally, we address the issue of whether our measure of monetary policy surprises is truly exogenous. Romer and Romer (2000) have argued that news on FOMC dates could reflect the Fed's superior ability to forecast economic activity, as opposed to orthogonal policy surprises. To support this contention, they show that Federal Reserve "Greenbook" forecasts consistently outperform private sector forecasts. We address this issue in two ways. First, following Barakchian and Crowe (2013) we construct a measure of the Fed's "private information" by taking the difference between the Greenbook forecast and the private sector forecast of the same economic activity variable, where we use the Blue Chip Economic Indicators Survey to measure the latter. Then we regress our measure of monetary policy surprises on this private information measure to determine the variation in the former explained by the latter. Second, we then use the residuals from this regression to construct a new measure of policy surprises that eliminates the component that may be due to the Fed's private information. We then repeat the VAR analysis using this "purified" measure of policy surprises. We note that the Greenbook data is only available through 2007 and only for scheduled FOMC dates. We thus limit the sample used for the identification of monetary policy shock accordingly.

TABLE 4—EFFECTS OF PRIVATE INFORMATION ON
TIGHT WINDOW MONETARY POLICY SURPRISE (1991–2007)

Variables	FF1 (1)	FF4 (2)	ED4 (3)
π	0.0227** (2.161)	0.0145** (2.109)	0.0152 (1.611)
dy	0.0166* (1.724)	0.0209*** (3.077)	0.0256*** (3.072)
$\Delta\pi$	−0.0289** (−2.387)	−0.0178* (−1.925)	−0.0185 (−1.528)
Δdy	−0.00663 (−1.309)	−0.00755* (−1.881)	−0.00627 (−1.033)
Observations	141	141	141
R^2	0.108	0.155	0.135
F -statistic	2.175	3.243	3.368
prob > F	0.0751	0.0141	0.0116

Note: Robust t -statistics in parentheses.

*** Significant at the 1 percent level.

** Significant at the 5 percent level.

* Significant at the 10 percent level.

Table 4 reports the results from regressions of surprises in interest rate futures on measures of the Fed's private information. The three dependent variables we consider are the surprises (on FOMC dates) in: the current month's fed funds future rate (FF1); the three month ahead fed funds future rate FF4; and the year ahead three month Eurodollar rate (ED4). The four independent variables include the Fed's private information for current inflation π , for current output growth dy , and for the change in each of these variables from the previous meeting, $\Delta\pi$ and Δdy .¹⁹ In each case, the private information variables are statistically significant but explain only a small fraction of the variation in the policy surprise. In particular, the private information variables account for only 15 percent of the surprise variation in our baseline instrument FF4 and slightly less of the surprise variation in the other instruments. Thus, we conclude that the Fed's private information cannot account for the overall variation in our policy surprises measures.

We next turn to the VAR analysis. We first “clean” our measure of policy surprises of the Fed's private measure by using the residuals from the regression for FF4 in Table 4 as our new instrument. In principle, these residuals capture variation in the surprise in FF4 on FOMC dates that is orthogonal to the Fed's private information. Figure 10 then reports the effect of a surprise monetary tightening using the “purified” measure of policy shocks. As one might expect, the effect of the surprise tightening is stronger than in the baseline: the fall in output and inflation is larger and the increases in spreads is greater. For example, private information that the economy is stronger than the public expects is likely to cause the Fed to raise rates more than expected, leading to procyclical movement in output and the policy indicator. By cleaning off the private information, it is possible to isolate the contractionary

¹⁹ Adding additional lags of these variables does not improve predictability.

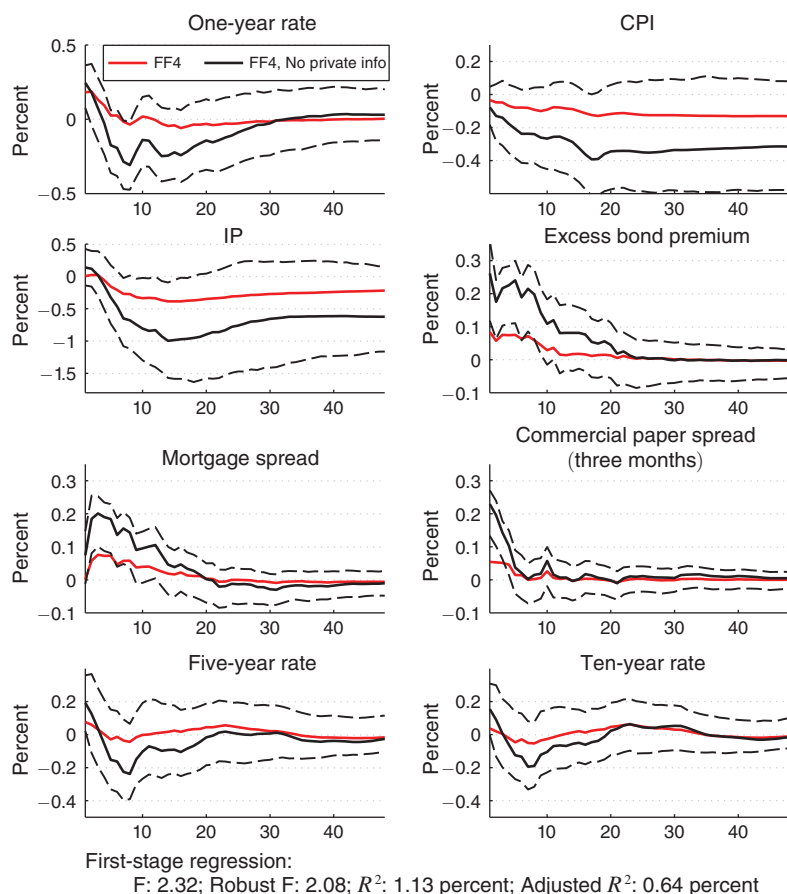


FIGURE 10. ONE-YEAR RATE SHOCK WITH INSTRUMENTS WITHOUT THE FED'S PRIVATE INFORMATION, 1979–2012

effects of the policy tightening. A caveat, however, having to limit the sample due to the availability of the Greenbook forecast data implies that in contrast to our baseline case we cannot confidently rule out a problem of weak instruments.²⁰

IV. Conclusion

Conventional VAR analysis of monetary shocks typically finds that relatively modest and transitory increases in short-term interest rates lead to significant contractions in output. We reexamine this issue using an external instruments identification approach borrowing insights from the high frequency identification literature of Kuttner (2001) and Gürkaynak, Sack, and Swanson (2005). The approach facilitates analyzing the joint response of financial and real variables to monetary

²⁰The large effects of the cleaned surprise measure is partly coming from our method of shock normalization. In particular, we normalize the impulse responses to reflect similar effects on the monetary policy indicator variable in the first three months. The cleaned surprise measure implies significantly smaller effects on the monetary policy indicator, and only slightly larger effects on prices, economic activity, and premia. As the normalization increases the monetary policy shock size, these responses also get amplified.

policy shocks and also allows for incorporating innovations in forward guidance into the measure of the shock. A key finding is that there is an enhanced movement in credit costs from the policy shock due to the response of term premia and credit costs. The response of credit costs is thus significantly larger than inspection of the short-rate response alone would suggest. These kinds of phenomena are absent in standard models of monetary policy transmission.

Overall, our results suggest a need to incorporate term premium and credit spread effects in the modeling of monetary policy transmission. One example of this approach, is Gertler and Karadi (2013) who extend a conventional sticky price monetary DSGE model to allow for limited participation in financial markets and limits to arbitrage. Within this framework a monetary tightening can produce increase in both term premia and credit spreads broadly consistent with the evidence in this paper. A byproduct of limited participation/limited arbitrage is that asset supplies affect asset prices, opening up a role for quantitative easing. There are other approaches as well worth pursuing.

REFERENCES

- Bagliano, Fabio C., and Carlo A. Favero.** 1999. "Information from financial markets and VAR measures of monetary policy." *European Economic Review* 43 (4–6): 825–37.
- Barakchian, S. Mahdi, and Christopher Crowe.** 2013. "Monetary policy matters: Evidence from new shocks data." *Journal of Monetary Economics* 60 (8): 950–66.
- Bernanke, Ben S., and Alan S. Blinder.** 1992. "The Federal Funds Rate and the Channels of Monetary Transmission." *American Economic Review* 82 (4): 901–21.
- Bernanke, Ben S., and Mark Gertler.** 1995. "Inside the Black Box: The Credit Channel of Monetary Policy Transmission." *Journal of Economic Perspectives* 90 (4): 27–48.
- Bernanke, Ben S., Mark Gertler, and Simon Gilchrist.** 1999. "The Financial Accelerator in a Quantitative Business Cycle Framework." In *Handbook of Macroeconomics*, Vol. 1, edited by J. B. Taylor and M. Woodford, 1341–93. Amsterdam: North Holland.
- Bernanke, Ben S., Vincent R. Reinhart, and Brian P. Sack.** 2004. "Monetary Policy Alternatives at the Zero Bound: An Empirical Assessment." *Brookings Papers on Economic Activity* 34 (2): 1–100.
- Boivin, Jean, Michael T. Kiley, and Frederic S. Mishkin.** 2010. "How has the monetary transmission mechanism evolved over time?" In *Handbook of Monetary Economics*, Vol. 3, edited by Benjamin M. Friedman and Michael Woodford, 369–422. Amsterdam: North Holland.
- Brunnermeier, Markus K., and Yuliy Sannikov.** 2011. "The I theory of money." http://scholar.princeton.edu/sites/default/files/i_theory_0.pdf.
- Campbell, Jeffrey R., Charles L. Evans, Jonas D. M. Fisher, Alejandro Justiniano.** 2012. "Macroeconomic effects of federal reserve forward guidance." *Brookings Papers on Economic Activity* 42 (1): 1–80.
- Christiano, Lawrence J., Martin Eichenbaum, and Charles L. Evans.** 1996. "The Effects of Monetary Policy Shocks: Evidence from the Flow of Funds." *Review of Economics and Statistics* 78 (1): 16–34.
- Christiano, Lawrence J., Martin Eichenbaum, and Charles L. Evans.** 2005. "Nominal rigidities and the dynamic effects of a shock to monetary policy." *Journal of Political Economy* 113 (1): 1–45.
- Christiano, Lawrence J., Roberto Motto, and Massimo Rostagno.** 2014. "Risk Shocks." *American Economic Review* 104 (1): 27–65.
- Clarida, Richard, Jordi Gali, and Mark Gertler.** 2000. "Monetary policy rules and macroeconomic stability: Evidence and some theory." *Quarterly Journal of Economics* 115 (1): 147–80.
- Cochrane, John H., and Monika Piazzesi.** 2002. "The Fed and Interest Rates—A High-Frequency Identification." *American Economic Review* 92 (2): 90–95.
- Drechsler, Itamar, Alexi Savov, and Philipp Schnabl.** 2014. "A Model of Monetary Policy and Risk Premia." National Bureau of Economic Research (NBER) Working Paper 20141.
- Faust, Jon, Eric T. Swanson, and Jonathan H. Wright.** 2004. "Identifying VARs based on high frequency futures data." *Journal of Monetary Economics* 51 (6): 1107–31.
- Gertler, Mark, and Peter Karadi.** 2011. "A model of unconventional monetary policy." *Journal of Monetary Economics* 58 (1): 17–34.

- Gertler, Mark, and Peter Karadi. 2013. "QE 1 vs. 2 vs. 3...: A Framework for Analyzing Large-Scale Asset Purchases as a Monetary Policy Tool." *International Journal of Central Banking* 90 (1): 5–53.
- Gertler, Mark, and Peter Karadi. 2015. "Monetary Policy Surprises, Credit Costs, and Economic Activity: Dataset." *American Economic Journal: Macroeconomics*. <http://dx.doi.org/10.1257/mac.20130329>.
- Gilchrist, Simon, and Egon Zakrajšek. 2012. "Credit spreads and business cycle fluctuations." *American Economic Review* 102 (4): 1692–1720.
- Gürkaynak, Refet S., Brian Sack, and Eric T. Swanson. 2005. "Do Actions Speak Louder than Words? The Response of Asset Prices to Monetary Policy Actions and Statements." *International Journal of Central Banking* 1 (1): 55–93.
- Gürkaynak, Refet S., Brian Sack, and Jonathan H. Wright. 2007. "The U.S. Treasury yield curve: 1961 to the present." *Journal of Monetary Economics* 54 (8): 2291–2304.
- Gürkaynak, Refet S., Brian Sack, and Jonathan H. Wright. 2010. "The TIPS Yield Curve and Inflation Compensation." *American Economic Journal: Macroeconomics* 2 (1): 70–92.
- Hamilton, James D. 2008. "Daily monetary policy shocks and new home sales." *Journal of Monetary Economics* 55 (7): 1171–90.
- Hamilton, James D. 2009. "Daily Changes in Fed Funds Futures Prices." *Journal of Money, Credit and Banking* 41 (4): 567–82.
- Hanson, Samuel G., and Jeremy C. Stein. 2012. "Monetary policy and long-term real rates." Finance and Economics Discussion Series (FEDS) Discussion Paper 2012–46.
- Kuttner, Kenneth N. 2001. "Monetary policy surprises and interest rates: Evidence from the Fed funds futures market." *Journal of Monetary Economics* 47 (3): 523–44.
- Mertens, Karel, and Morten O. Ravn. 2013. "The Dynamic Effects of Personal and Corporate Income Tax Changes in the United States." *American Economic Review* 103 (4): 1212–47.
- Nakamura, Emi, and Jón Steinsson. 2013. "High Frequency Identification of Monetary Non-Neutrality." National Bureau of Economic Research (NBER) Working Paper 19260.
- Piazzesi, Monika, and Eric T. Swanson. 2008. "Futures prices as risk-adjusted forecasts of monetary policy." *Journal of Monetary Economics* 550 (4): 677–91.
- Romer, Christina D., and David H. Romer. 2000. "Federal Reserve Information and the Behavior of Interest Rates." *American Economic Review* 90 (3): 429–57.
- Romer, Christina D. and David H. Romer. 2004. "A New Measure of Monetary Shocks: Derivation and Implications." *American Economic Review* 94 (3): 1055–84.
- Rudebusch, Glenn D. 1998. "Do Measures of Monetary Policy in a VAR Make Sense?" *International Economic Review* 39 (4): 907–31.
- Smets, Frank, and Rafael Wouters. 2007. "Shocks and Frictions in US Business Cycles: A Bayesian DSGE Approach." *American Economic Review* 970 (3): 586–606.
- Stock, James H., and Mark W. Watson. 2012. "Disentangling the channels of the 2007–09 recession." *Brookings Papers on Economic Activity* 42 (1): 81–135.
- Stock, James H., Jonathan H. Wright, and Motohiro Yogo. 2002. "A survey of weak instruments and weak identification in generalized method of moments." *Journal of Business & Economic Statistics* 20 (4): 518–29.
- Swanson, Eric T., and John C. Williams. 2014. "Measuring the Effect of the Zero Lower Bound on Medium- and Longer-Term Interest Rates." *American Economic Review* 104 (10): 3154–85.

This article has been cited by:

1. Meihua Jiang, Yuzhe Huang. 2023. Is forward guidance an effective policy: A time-varying analysis. *Finance Research Letters* **58**, 104486. [[Crossref](#)]
2. Pedro Gomis-Porqueras, Shuddhasattwa Rafiq, Wenying Yao. 2023. The impact of forward guidance and large-scale asset purchase programs on commodity markets. *Studies in Nonlinear Dynamics & Econometrics* **27**:4, 519-551. [[Crossref](#)]
3. BENJAMIN BECKERS, KERSTIN BERNOTH. 2023. Monetary Policy and Mispricing in Stock Markets. *Journal of Money, Credit and Banking* **81**. . [[Crossref](#)]
4. Fergus Cumming, Paul Hubert. 2023. The Distribution of Households' Indebtedness and the Transmission of Monetary Policy. *Review of Economics and Statistics* 1-10. [[Crossref](#)]
5. Matthew Read. 2023. Estimating the Effects of Monetary Policy in Australia Using Sign-restricted Structural Vector Autoregressions*. *Economic Record* . [[Crossref](#)]
6. Martin Feldkircher, Helene Schuberth. 2023. Understanding Monetary Spillovers in Highly Integrated Regions: The Case of Europe*. *Oxford Bulletin of Economics and Statistics* **85**:4, 859-893. [[Crossref](#)]
7. Samer Adra, Elie Menassa. 2023. Uncertainty and corporate investments in response to the Fed's dual shocks. *Financial Review* **58**:3, 463-484. [[Crossref](#)]
8. Sebastian K. R  th, Wouter Van der Veken. 2023. Monetary policy and exchange rate anomalies in set-identified SVARs: Revisited. *Journal of Applied Econometrics* . [[Crossref](#)]
9. Wenting Liao, Jun Ma, Chengsi Zhang. 2023. Identifying exchange rate effects and spillovers of US monetary policy shocks in the presence of time-varying instrument relevance. *Journal of Applied Econometrics* **1**. . [[Crossref](#)]
10. Lukas Hoesch, Barbara Rossi, Tatevik Sekhposyan. 2023. Has the Information Channel of Monetary Policy Disappeared? Revisiting the Empirical Evidence. *American Economic Journal: Macroeconomics* **15**:3, 355-387. [[Abstract](#)] [[View PDF article](#)] [[PDF with links](#)]
11. Stylianos Tsiaras. 2023. Asset Purchases, Limited Asset Markets Participation and Inequality. *Journal of Economic Dynamics and Control* **14**, 104721. [[Crossref](#)]
12. Jo  o F. Gomes, Marco Grotteria, Jessica A. Wachter. 2023. Foreseen Risks. *Journal of Economic Theory* 105706. [[Crossref](#)]
13. Nuri Volkan KAYA  ET  N. 2023. ELECTIONS AND STOCK MARKET RETURNS: EVIDENCE FROM BORSA ISTANBUL. *Journal of Research in Business* **8**:1, 20-40. [[Crossref](#)]
14. Martin Hodula, Jan Libich. 2023. Has monetary policy fueled the rise in shadow banking?. *Economic Modelling* **123**, 106278. [[Crossref](#)]
15. Michael Funke, Xiang Li, Doudou Zhong. 2023. Household indebtedness, financial frictions and the transmission of monetary policy to consumption: Evidence from China. *Emerging Markets Review* **55**, 100974. [[Crossref](#)]
16. Akito Matsumoto, Andrea Pescatori, Xueliang Wang. 2023. Commodity prices and global economic activity. *Japan and the World Economy* **66**, 101177. [[Crossref](#)]
17. Yang Yang, Jiqiang Zhang, Sanpan Chen. 2023. Information effects of monetary policy announcements on oil price. *Journal of Commodity Markets* **30**, 100268. [[Crossref](#)]
18. Clara De Luigi, Martin Feldkircher, Philipp Poyntner, Helene Schuberth. 2023. Quantitative Easing and Wealth Inequality: The Asset Price Channel*. *Oxford Bulletin of Economics and Statistics* **85**:3, 638-670. [[Crossref](#)]

19. Jörg Breitung, Ralf Brüggemann. 2023. Projection Estimators for Structural Impulse Responses*. *Oxford Bulletin of Economics and Statistics* **79**. . [[Crossref](#)]
20. Carlos Alba, Gabriel Cuadra, Raul Ibarra. 2023. The effects of central bank extraordinary measures on financial conditions: Evidence from Mexico. *Applied Economics* **10**, 1-22. [[Crossref](#)]
21. Haroon Mumtaz, Konstantinos Theodoridis. 2023. THE FEDERAL RESERVE'S IMPLICIT INFLATION TARGET AND MACROECONOMIC DYNAMICS: AN SVAR ANALYSIS. *International Economic Review* **120**. . [[Crossref](#)]
22. Rashad Ahmed. 2023. Flights-to-safety and macroeconomic adjustment in emerging markets: The role of U.S. monetary policy. *Journal of International Money and Finance* **133**, 102827. [[Crossref](#)]
23. Ben S. Bernanke. 2023. Nobel Lecture: Banking, Credit, and Economic Fluctuations. *American Economic Review* **113**:5, 1143-1169. [[Abstract](#)] [[View PDF article](#)] [[PDF with links](#)]
24. Mark W. Watson. 2023. Comment. *NBER Macroeconomics Annual* **37**, 161-166. [[Crossref](#)]
25. Michael D. Bauer, Eric T. Swanson. 2023. A Reassessment of Monetary Policy Surprises and High-Frequency Identification. *NBER Macroeconomics Annual* **37**, 87-155. [[Crossref](#)]
26. Christian K. Wolf. 2023. Fiscal Stimulus and the Systematic Response of Monetary Policy. *AEA Papers and Proceedings* **113**, 388-393. [[Abstract](#)] [[View PDF article](#)] [[PDF with links](#)]
27. Giovanni Pellegrino, Efrem Castelnuovo, Giovanni Caggiano. 2023. UNCERTAINTY AND MONETARY POLICY DURING THE GREAT RECESSION. *International Economic Review* **64**:2, 577-606. [[Crossref](#)]
28. Harald Badinger, Stefan Schiman. 2023. Measuring Monetary Policy in the Euro Area Using SVARs with Residual Restrictions. *American Economic Journal: Macroeconomics* **15**:2, 279-305. [[Abstract](#)] [[View PDF article](#)] [[PDF with links](#)]
29. Daniel Gründler, Eric Mayer, Johann Scharler. 2023. Monetary Policy Announcements, Information Shocks, and Exchange Rate Dynamics. *Open Economies Review* **34**:2, 341-369. [[Crossref](#)]
30. Juan Morelli, Matias Moretti. 2023. Information Frictions, Reputation, and Sovereign Spreads. *Journal of Political Economy* **86**. . [[Crossref](#)]
31. Michael D. Bauer, Eric T. Swanson. 2023. An Alternative Explanation for the “Fed Information Effect”. *American Economic Review* **113**:3, 664-700. [[Abstract](#)] [[View PDF article](#)] [[PDF with links](#)]
32. Zhenzhu Chen, Li Li, Changhua Yu. 2023. Monetary policy shocks and the signaling channel of monetary policy in China. *China Economic Quarterly International* **3**:1, 46-59. [[Crossref](#)]
33. Robin Döttling, Lev Ratnovski. 2023. Monetary policy and intangible investment. *Journal of Monetary Economics* **134**, 53-72. [[Crossref](#)]
34. Katarzyna Budnik, Gerhard Rünstler. 2023. Identifying structural VARs from sparse narrative instruments: Dynamic effects of US macroprudential policies. *Journal of Applied Econometrics* **38**:2, 186-201. [[Crossref](#)]
35. HAROON MUMTAZ, KATERINA PETROVA. 2023. Changing Impact of Shocks: A Time-Varying Proxy SVAR Approach. *Journal of Money, Credit and Banking* **55**:2-3, 635-654. [[Crossref](#)]
36. Luccas Assis Attílio, João Ricardo Faria, Mauro Rodrigues. 2023. Does monetary policy impact CO2 emissions? A GVAR analysis. *Energy Economics* **119**, 106559. [[Crossref](#)]

37. Alexander Jung. 2023. Are monetary policy shocks causal to bank health? Evidence from the euro area. *Journal of Macroeconomics* **75**, 103494. [[Crossref](#)]
38. Amadeu DaSilva, Mira Farka. 2023. Business cycles, stock returns and the transmission channels of conventional and unconventional monetary policy. *Applied Economics Letters* **79**, 1-9. [[Crossref](#)]
39. ANNA SAMARINA, ANH D.M. NGUYEN. 2023. Does Monetary Policy Affect Income Inequality in the Euro Area?. *Journal of Money, Credit and Banking* **89**. . [[Crossref](#)]
40. Abhishek Kumar, Sushanta Mallick, Madhusudan Mohanty, Fabrizio Zampolli. 2023. Market Volatility, Monetary Policy and the Term Premium. *Oxford Bulletin of Economics and Statistics* **85**:1, 208-237. [[Crossref](#)]
41. KNUT ARE AASTVEIT, BRUNO ALBUQUERQUE, ANDRÉ K. ANUNDSEN. 2023. Changing Supply Elasticities and Regional Housing Booms. *Journal of Money, Credit and Banking* **72**. . [[Crossref](#)]
42. Martin Geiger, Daniel Gründler, Johann Scharler. 2023. Monetary policy shocks and consumer expectations in the euro area. *Journal of International Economics* **140**, 103708. [[Crossref](#)]
43. James McNeil. 2023. Monetary policy and the term structure of inflation expectations with information frictions. *Journal of Economic Dynamics and Control* **146**, 104588. [[Crossref](#)]
44. John Ammer, John Rogers, Gang Wang, Yang Yu. 2023. Chinese Asset Managers' Monetary Policy Forecasts and Fund Performance. *Management Science* **69**:1, 598-616. [[Crossref](#)]
45. Florian Gerth. An anatomy of interest rates: How do interest rates really work? . [[Crossref](#)]
46. Andi Duqi. The Role of Banks in Promoting Post-disaster Economic Growth 59-98. [[Crossref](#)]
47. Thanh Phuc Nguyen, Tho Ngoc Tran, Thi Thu Hong Dinh, Tri M. Hoang, Trang Duong Thi Thuy. 2022. Drivers of climate change in selected emerging countries: the ecological effects of monetary restrictions and expansions. *Cogent Economics & Finance* **10**:1. . [[Crossref](#)]
48. Woon Wook Jang. 2022. Monetary policy shocks identified using the entire yield curve: an alternative approach. *Applied Economics Letters* **29**:21, 2020-2031. [[Crossref](#)]
49. Carlos Arteta, Steven Kamin, Ruch Franz Ulrich. How do Rising U.S. Interest Rates Affect Emerging and Developing Economies? It Depends **12**, . [[Crossref](#)]
50. François-Éric Racicot, Raymond Théoret. 2022. Tracking market and non-traditional sources of risks in procyclical and countercyclical hedge fund strategies under extreme scenarios: a nonlinear VAR approach. *Financial Innovation* **8**:1. . [[Crossref](#)]
51. Xiao-Li Gong, Jin-Yan Lu, Xiong Xiong, Wei Zhang. 2022. Higher-order dynamic effects of uncertainty risk under thick-tailed stochastic volatility. *Financial Innovation* **8**:1. . [[Crossref](#)]
52. BERRAK BAHADIR, DORA GICHEVA. 2022. Macroeconomic Implications of Student Debt: A State-Level Analysis. *Journal of Money, Credit and Banking* **54**:8, 2273-2300. [[Crossref](#)]
53. JULIAN DI GIOVANNI, GALINA HALE. 2022. Stock Market Spillovers via the Global Production Network: Transmission of U.S. Monetary Policy. *The Journal of Finance* **77**:6, 3373-3421. [[Crossref](#)]
54. Jérôme Creel, Mehdi El Herradi. 2022. Income inequality and monetary policy in the euro area. *International Journal of Finance & Economics* **82**. . [[Crossref](#)]
55. Jochen Güntner. 2022. Central bank information and private-sector expectations. *Journal of Forecasting* **41**:7, 1372-1385. [[Crossref](#)]
56. Wongi Kim, Kyunghun Kim. 2022. Effect of news and noise shocks of US monetary policy on economic fluctuations in emerging market economies. *Canadian Journal of Economics/Revue canadienne d'économie* **55**:4, 1862-1893. [[Crossref](#)]

57. Omar Abdulla Mohammed Hajeej. 2022. #####. *Tikrit Journal of Administrative and Economic Sciences* 18:59, 2, 418-432. [[Crossref](#)]
58. Carsten Jentsch, Kurt G. Lunsford. 2022. Asymptotically Valid Bootstrap Inference for Proxy SVARs. *Journal of Business & Economic Statistics* 40:4, 1876-1891. [[Crossref](#)]
59. Fernando Ballabriga, Karen Davtyan. 2022. Distributional impact of monetary policy in the UK: from conventional to unconventional policy. *Journal of Economic Policy Reform* 25:4, 435-450. [[Crossref](#)]
60. Juan-Francisco Albert, Nerea Gómez-Fernández. 2022. Monetary policy and the redistribution of net worth in the U.S. *Journal of Economic Policy Reform* 25:4, 420-434. [[Crossref](#)]
61. Knut Are Aastveit, André K. Anundsen. 2022. Asymmetric Effects of Monetary Policy in Regional Housing Markets. *American Economic Journal: Macroeconomics* 14:4, 499-529. [[Abstract](#)] [[View PDF article](#)] [[PDF with links](#)]
62. John V. Leahy, Aditi Thapar. 2022. Age Structure and the Impact of Monetary Policy. *American Economic Journal: Macroeconomics* 14:4, 136-173. [[Abstract](#)] [[View PDF article](#)] [[PDF with links](#)]
63. Mattias Almgren, José-Elías Gallegos, John Kramer, Ricardo Lima. 2022. Monetary Policy and Liquidity Constraints: Evidence from the Euro Area. *American Economic Journal: Macroeconomics* 14:4, 309-340. [[Abstract](#)] [[View PDF article](#)] [[PDF with links](#)]
64. Annalisa Ferrando, Alexander Popov, Gregory F. Udell. 2022. Unconventional monetary policy, funding expectations, and firm decisions. *European Economic Review* 149, 104268. [[Crossref](#)]
65. Martin Weale, Tomasz Wieladek. 2022. Financial effects of QE and conventional monetary policy compared. *Journal of International Money and Finance* 127, 102673. [[Crossref](#)]
66. Mi Zhang, Ahmet Sensoy, Feiyang Cheng, Xuankai Zhao. 2022. Three channels of monetary policy international transmission: Identifying spillover effects from the US to China. *Research in International Business and Finance* 61, 101670. [[Crossref](#)]
67. Jiarui Zhang, Yingying Shi. 2022. The resource reallocation effect of monetary policy. *International Journal of Finance & Economics* 122. . [[Crossref](#)]
68. Yanfeng Wei, Xiaoying Guo. 2022. The impact of oil supply shocks on real economic activity: new evidence based on the proxy SVARs. *Applied Economics* 54:44, 5035-5049. [[Crossref](#)]
69. Niklas Amberg, Thomas Jansson, Mathias Klein, Anna Rogantini Picco. 2022. Five Facts about the Distributional Income Effects of Monetary Policy Shocks. *American Economic Review: Insights* 4:3, 289-304. [[Abstract](#)] [[View PDF article](#)] [[PDF with links](#)]
70. Pierpaolo Benigno, Paolo Canofari, Giovanni Di Bartolomeo, Marcello Messori. 2022. The European Monetary Policy Responses During the Pandemic Crisis. *Open Economies Review* 33:4, 657-675. [[Crossref](#)]
71. Maciej Stefański. 2022. Macroeconomic effects and transmission channels of quantitative easing. *Economic Modelling* 114, 105943. [[Crossref](#)]
72. Samer Adra, Elie Menassa. 2022. The Fed's dual shocks and the housing market. *Economics Letters* 218, 110730. [[Crossref](#)]
73. Samer Adra, Leonidas G. Barbopoulos. 2022. Monetary shocks and the analyst coverage of the firm. *Economics Letters* 218, 110776. [[Crossref](#)]
74. MICHAEL ELLINGTON. 2022. The Empirical Relevance of the Shadow Rate and the Zero Lower Bound. *Journal of Money, Credit and Banking* 54:6, 1605-1635. [[Crossref](#)]

75. Hasan Dinger, Serhat Yüksel. Monetary Policy Operations of Central Banks in the E7 Economies 959-978. [[Crossref](#)]
76. Petr Makovský. 2022. Are Term Premiums Predictable in Central European Countries? The Forward Rates Agreements (FRA) Application. *Eastern European Economics* 36, 1-32. [[Crossref](#)]
77. Anastasios Evgenidis, Anastasios G. Malliaris. 2022. Monetary policy, financial shocks and economic activity. *Review of Quantitative Finance and Accounting* 59:2, 429-456. [[Crossref](#)]
78. Mikkel Plagborg-Møller, Christian K. Wolf. 2022. Instrumental Variable Identification of Dynamic Variance Decompositions. *Journal of Political Economy* 130:8, 2164-2202. [[Crossref](#)]
79. REFET GÜRKAYNAK, HATİCE GÖKÇE KARASOY-CAN, SANG SEOK LEE. 2022. Stock Market's Assessment of Monetary Policy Transmission: The Cash Flow Effect. *The Journal of Finance* 77:4, 2375-2421. [[Crossref](#)]
80. YIFEI WANG, TONI M. WHITED, YUFENG WU, KAIRONG XIAO. 2022. Bank Market Power and Monetary Policy Transmission: Evidence from a Structural Estimation. *The Journal of Finance* 77:4, 2093-2141. [[Crossref](#)]
81. Nihar Shah. 2022. Doubly heterogeneous monetary spillovers. *International Finance* 25:2, 126-150. [[Crossref](#)]
82. SASKIA TER ELLEN, VEGARD H. LARSEN, LEIF ANDERS THORSRUD. 2022. Narrative Monetary Policy Surprises and the Media. *Journal of Money, Credit and Banking* 54:5, 1525-1549. [[Crossref](#)]
83. Robin Braun, Ralf Brüggemann. 2022. Identification of SVAR Models by Combining Sign Restrictions With External Instruments. *Journal of Business & Economic Statistics* 1-38. [[Crossref](#)]
84. Ursel Baumann, David Lodge, Mirela S. Miescu. 2022. Global growth on life support? The contributions of fiscal and monetary policy since the global financial crisis. *International Journal of Finance & Economics* 18. . [[Crossref](#)]
85. Alexander Chudik, Georgios Georgiadis. 2022. Estimation of Impulse Response Functions When Shocks Are Observed at a Higher Frequency Than Outcome Variables. *Journal of Business & Economic Statistics* 40:3, 965-979. [[Crossref](#)]
86. Helmut Lütkepohl, Thore Schlaak. 2022. Heteroscedastic Proxy Vector Autoregressions. *Journal of Business & Economic Statistics* 40:3, 1268-1281. [[Crossref](#)]
87. Mark Kerstenfischer. 2022. Information effects of euro area monetary policy. *Economics Letters* 216, 110570. [[Crossref](#)]
88. Jonathan Benchimol, Inon Gamrasni, Michael Kahn, Sigal Ribon, Yossi Saadon, Noam Ben-Ze'ev, Asaf Segal, Yitzchak Shizgal. 2022. The interaction between domestic monetary policy and macroprudential policy in Israel. *Economic Modelling* 112, 105872. [[Crossref](#)]
89. Carlo Altavilla, Lorenzo Burlon, Mariassunta Giannetti, Sarah Holton. 2022. Is there a zero lower bound? The effects of negative policy rates on banks and firms. *Journal of Financial Economics* 144:3, 885-907. [[Crossref](#)]
90. Seungyoon Lee, Jongwook Park. 2022. Identifying monetary policy shocks using economic forecasts in Korea. *Economic Modelling* 111, 105803. [[Crossref](#)]
91. Robert Jarrow, Sujam Lamichhane. 2022. Risk premia, asset price bubbles, and monetary policy. *Journal of Financial Stability* 60, 101005. [[Crossref](#)]
92. Yohei Yamamoto, Naoko Hara. 2022. Identifying factor-augmented vector autoregression models via changes in shock variances. *Journal of Applied Econometrics* 37:4, 722-745. [[Crossref](#)]

93. Andrea Carriero, Francesco Corsello, Massimiliano Marcellino. 2022. The global component of inflation volatility. *Journal of Applied Econometrics* 37:4, 700-721. [[Crossref](#)]
94. Stéphane Lhuissier, Urszula Szczerbowicz. 2022. Monetary Policy and Corporate Debt Structure*. *Oxford Bulletin of Economics and Statistics* 84:3, 497-515. [[Crossref](#)]
95. Regis Barnichon, Christian Matthes, Alexander Ziegenbein. 2022. Are the Effects of Financial Market Disruptions Big or Small?. *The Review of Economics and Statistics* 104:3, 557-570. [[Crossref](#)]
96. Martin Feldkircher, Kazuhiko Kakamu. 2022. How does monetary policy affect income inequality in Japan? Evidence from grouped data. *Empirical Economics* 62:5, 2307-2327. [[Crossref](#)]
97. Zehao Li. 2022. Financial intermediary leverage and monetary policy transmission. *European Economic Review* 144, 104080. [[Crossref](#)]
98. Masashige Hamano, Francesco Zanetti. 2022. Monetary policy, firm heterogeneity, and product variety. *European Economic Review* 144, 104089. [[Crossref](#)]
99. Max Breitenlechner, Martin Geiger, Johann Scharler. 2022. Bank liquidity and the propagation of uncertainty in the U.S. *Finance Research Letters* 46, 102467. [[Crossref](#)]
100. Olajide Omowonuola Oyadeyi. 2022. A systematic and non-systematic approach to monetary policy shocks and monetary transmission process in Nigeria. *Journal of Economics and International Finance* 14:2, 23-31. [[Crossref](#)]
101. Fabio Canetg, Daniel Kaufmann. 2022. Overnight rate and signalling effects of central bank bills. *European Economic Review* 143, 104060. [[Crossref](#)]
102. Nicolas Soenen, Rudi Vander Venet. 2022. ECB monetary policy and bank default risk#. *Journal of International Money and Finance* 122, 102571. [[Crossref](#)]
103. Samer Adra. 2022. Monetary policy and anxious presidents: The effects of monetary shocks on presidential job approval. *Politics & Policy* 50:2, 244-273. [[Crossref](#)]
104. FRANCESCO BIANCHI, MARTIN LETTAU, SYDNEY C. LUDVIGSON. 2022. Monetary Policy and Asset Valuation. *The Journal of Finance* 77:2, 967-1017. [[Crossref](#)]
105. Martin Kliem, Alexander Meyer-Gohde. 2022. (Un)expected monetary policy shocks and term premia. *Journal of Applied Econometrics* 37:3, 477-499. [[Crossref](#)]
106. Samer Adra, Elie Menassa. 2022. Central Bank Information Shocks, Value Gains, and Value Crashes. *Journal of Behavioral Finance* 118, 1-15. [[Crossref](#)]
107. Martin Eichenbaum, Sergio Rebelo, Arlene Wong. 2022. State-Dependent Effects of Monetary Policy: The Refinancing Channel. *American Economic Review* 112:3, 721-761. [[Abstract](#)] [[View PDF article](#)] [[PDF with links](#)]
108. Olatunji Abdul Shobande. 2022. Is climate change a monetary phenomenon? Evidence from time series analysis. *International Journal of Sustainable Development & World Ecology* 29:2, 99-111. [[Crossref](#)]
109. Nina Boyarchenko, Giovanni Favara, Moritz Schularick. 2022. Financial Stability Considerations for Monetary Policy: Empirical Evidence and Challenges. *Finance and Economics Discussion Series* 2022:005, 1-33. [[Crossref](#)]
110. Francesco Simone Lucidi, Willi Semmler. 2022. Supervisory shocks to banks' credit standards and their macroeconomic impact. *Journal of Financial Stability* 58, 100966. [[Crossref](#)]
111. Stefano Fasani, Haroon Mumtaz, Lorenza Rossi. 2022. Monetary policy uncertainty and firm dynamics. *Review of Economic Dynamics* 191. . [[Crossref](#)]

112. TONY ZHANG. 2022. Monetary Policy Spillovers through Invoicing Currencies. *The Journal of Finance* 77:1, 129-161. [[Crossref](#)]
113. Sam Ouliaris, Adrian Pagan. 2022. Three Basic Issues that Arise when Using Informational Restrictions in SVARs*. *Oxford Bulletin of Economics and Statistics* 84:1, 1-20. [[Crossref](#)]
114. Jongrim Ha, Seohyun Lee, Inhwan So. 2022. The Impact of Uncertainty Shocks: Evidence from Geopolitical Swings on the Korean Peninsula*. *Oxford Bulletin of Economics and Statistics* 84:1, 21-56. [[Crossref](#)]
115. Nuno Coimbra, Daisoon Kim, Hélène Rey. 2022. Central Bank Policy and the concentration of risk: Empirical estimates. *Journal of Monetary Economics* 125, 182-198. [[Crossref](#)]
116. Martin Bruns, Helmut Lutkepohl. 2022. Comparison of local projection estimators for proxy vector autoregressions. *Journal of Economic Dynamics and Control* 134, 104277. [[Crossref](#)]
117. Ralf Brüggemann, Robin Braun. 2022. Identification of SVAR models by combining sign restrictions with external instruments. *SSRN Electronic Journal* 34. . [[Crossref](#)]
118. Leonidas G. Barbopoulos, Samer Adra, Anthony Saunders. 2022. The Real Effects of Fed Information Shocks on the M&A Market. *SSRN Electronic Journal* . [[Crossref](#)]
119. Callum Jones, Mariano Kulish, Daniel M. Rees. 2022. International spillovers of forward guidance shocks. *Journal of Applied Econometrics* 37:1, 131-160. [[Crossref](#)]
120. Massimo Guidolin, Valentina Massagli, Manuela Pedio. 2021. Does the cost of private debt respond to monetary policy? Heteroskedasticity-based identification in a model with regimes. *The European Journal of Finance* 27:18, 1804-1833. [[Crossref](#)]
121. Filipp Prokopev. 2021. Balance Sheet Channel of Monetary Policy: Evidence from Credit Spreads of Russian Firms. *Russian Journal of Money and Finance* 80:4, 3-30. [[Crossref](#)]
122. Hans-Martin Beyer, Bodo Herzog. 2021. Supply Chain Finance: Cost–Benefit Differentials under Reverse Factoring with Extended Payment Terms. *International Journal of Financial Studies* 9:4, 59. [[Crossref](#)]
123. Miguel-Ángel Galindo-Martín, María-Soledad Castaño-Martínez, María-Teresa Méndez-Picazo. 2021. Effects of the pandemic crisis on entrepreneurship and sustainable development. *Journal of Business Research* 137, 345-353. [[Crossref](#)]
124. Manfred M. Fischer, Florian Huber, Michael Pfarrhofer, Petra Stauffer-Steinnocher. 2021. The Dynamic Impact of Monetary Policy on Regional Housing Prices in the United States. *Real Estate Economics* 49:4, 1039-1068. [[Crossref](#)]
125. Sami Alpanda, Eleonora Granziera, Sarah Zubairy. 2021. State dependence of monetary policy across business, credit and interest rate cycles. *European Economic Review* 140, 103936. [[Crossref](#)]
126. Aktham Maghyreh, Hussein Abdoh. 2021. The effect of structural oil shocks on bank systemic risk in the GCC countries. *Energy Economics* 103, 105568. [[Crossref](#)]
127. Georgios Georgiadis, Ben Schumann. 2021. Dominant-currency pricing and the global output spillovers from US dollar appreciation. *Journal of International Economics* 133, 103537. [[Crossref](#)]
128. Jongrim Ha. 2021. Financial market spillovers of U.S. monetary policy shocks. *Review of International Economics* 29:5, 1221-1274. [[Crossref](#)]
129. Julian Kozlowski. 2021. Long-Term Finance and Investment with Frictional Asset Markets. *American Economic Journal: Macroeconomics* 13:4, 411-448. [[Abstract](#)] [[View PDF article](#)] [[PDF with links](#)]
130. Darja Milic. 2021. The impact of non-banking financial institutions on monetary policy transmission in Euro area. *Empirical Economics* 61:4, 1779-1817. [[Crossref](#)]

131. Martin Blomhoff Holm, Pascal Paul, Andreas Tischbirek. 2021. The Transmission of Monetary Policy under the Microscope. *Journal of Political Economy* **129**:10, 2861-2904. [[Crossref](#)]
132. Samer Adra, Elie Menassa. 2021. Monetary policy and information production in the secondary market. *Economics Letters* **207**, 110044. [[Crossref](#)]
133. Niko Hauzenberger, Michael Pfarrhofer. 2021. Bayesian State-Space Modeling for Analyzing Heterogeneous Network Effects of US Monetary Policy*. *The Scandinavian Journal of Economics* **123**:4, 1261-1291. [[Crossref](#)]
134. Maximilian Böck, Martin Feldkircher, Pierre L. Siklos. 2021. International Effects of Euro Area Forward Guidance*. *Oxford Bulletin of Economics and Statistics* **83**:5, 1066-1110. [[Crossref](#)]
135. Pavel S. Kapinos. 2021. Monetary policy news and systemic risk at the zero lower bound. *International Journal of Finance & Economics* **26**:4, 4932-4945. [[Crossref](#)]
136. Oguzhan Cepni, Rangan Gupta, Qiang Ji. 2021. Sentiment Regimes and Reaction of Stock Markets to Conventional and Unconventional Monetary Policies: Evidence from OECD Countries. *Journal of Behavioral Finance* **33**, 1-17. [[Crossref](#)]
137. David Berger, Konstantin Milbradt, Fabrice Tourre, Joseph Vavra. 2021. Mortgage Prepayment and Path-Dependent Effects of Monetary Policy. *American Economic Review* **111**:9, 2829-2878. [[Abstract](#)] [[View PDF article](#)] [[PDF with links](#)]
138. Max Breitenlechner, Johann Scharler. 2021. Monetary policy announcements and bank lending: Do banks' refinancing markets matter?. *Economic Modelling* **102**, 105559. [[Crossref](#)]
139. KIMBERLY A. BERG, CHADWICK C. CURTIS, STEVEN LUGAUER, NELSON C. MARK. 2021. Demographics and Monetary Policy Shocks. *Journal of Money, Credit and Banking* **53**:6, 1229-1266. [[Crossref](#)]
140. Martin Baumgärtner, Jens Klose. 2021. Why central banks announcing liquidity injections is more effective than forward guidance. *International Finance* **24**:2, 236-256. [[Crossref](#)]
141. Giovanni Pellegrino, Federico Ravenna, Gabriel Züllig. 2021. The Impact of Pessimistic Expectations on the Effects of COVID-19-Induced Uncertainty in the Euro Area*. *Oxford Bulletin of Economics and Statistics* **83**:4, 841-869. [[Crossref](#)]
142. Silvia Miranda-Agrippino, Giovanni Ricco. 2021. The Transmission of Monetary Policy Shocks. *American Economic Journal: Macroeconomics* **13**:3, 74-107. [[Abstract](#)] [[View PDF article](#)] [[PDF with links](#)]
143. Giovanni Pellegrino. 2021. Uncertainty and monetary policy in the US: A journey into nonlinear territory. *Economic Inquiry* **59**:3, 1106-1128. [[Crossref](#)]
144. V. A. Bannikova, A. A. Pestova. 2021. The effects of monetary shocks on inflation: High-frequency approach. *Voprosy Ekonomiki* :6, 47-76. [[Crossref](#)]
145. Mongi Arfaoui, Aymen Ben Rejeb. 2021. Modeling the volatility of DJIM equity indices: a fundamental analysis using quantile regression. *International Journal of Islamic and Middle Eastern Finance and Management* **14**:3, 482-505. [[Crossref](#)]
146. Markus Brunnermeier, Darius Palia, Karthik A. Sastry, Christopher A. Sims. 2021. Feedbacks: Financial Markets and Economic Activity. *American Economic Review* **111**:6, 1845-1879. [[Abstract](#)] [[View PDF article](#)] [[PDF with links](#)]
147. Michiel De Pooter, Giovanni Favara, Michele Modugno, Jason Wu. 2021. Reprint: Monetary policy uncertainty and monetary policy surprises. *Journal of International Money and Finance* **114**, 102401. [[Crossref](#)]

148. Etsuro Shioji. 2021. Pass-through of oil supply shocks to domestic gasoline prices: evidence from daily data. *Energy Economics* **98**, 105214. [[Crossref](#)]
149. KENT DANIEL, LORENZO GARLAPPI, KAIRONG XIAO. 2021. Monetary Policy and Reaching for Income. *The Journal of Finance* **76**:3, 1145-1193. [[Crossref](#)]
150. M. I. Stolbov. 2021. The global financial cycle: Notion, empirical evidence and policy implications. *Voprosy Ekonomiki* :5, 43-56. [[Crossref](#)]
151. Ralf Fendel, Frederik Neugebauer, Mustafa Kilinc. 2021. ECB's communication and the yield curve: core versus periphery effects. *Applied Economics Letters* **28**:8, 630-634. [[Crossref](#)]
152. Refet S. Gürkaynak, A. Hakan Kara, Burçin Kısacıkoğlu, Sang Seok Lee. 2021. Monetary policy surprises and exchange rate behavior. *Journal of International Economics* **130**, 103443. [[Crossref](#)]
153. Ethan Ilzetzki, Keyu Jin. 2021. The puzzling change in the international transmission of U.S. macroeconomic policy shocks. *Journal of International Economics* **130**, 103444. [[Crossref](#)]
154. Diego R. Känzig. 2021. The Macroeconomic Effects of Oil Supply News: Evidence from OPEC Announcements. *American Economic Review* **111**:4, 1092-1125. [[Abstract](#)] [[View PDF article](#)] [[PDF with links](#)]
155. Ralph Luetticke. 2021. Transmission of Monetary Policy with Heterogeneity in Household Portfolios. *American Economic Journal: Macroeconomics* **13**:2, 1-25. [[Abstract](#)] [[View PDF article](#)] [[PDF with links](#)]
156. Cosmas Dery, Apostolos Serletis. 2021. The Relative Importance of Monetary Policy, Uncertainty, and Financial Shocks. *Open Economies Review* **32**:2, 311-333. [[Crossref](#)]
157. Michiel De Pooter, Giovanni Favara, Michele Modugno, Jason Wu. 2021. Monetary policy uncertainty and monetary policy surprises. *Journal of International Money and Finance* **112**, 102323. [[Crossref](#)]
158. . Asset Purchases in Emerging Markets: Unconventional Policies, Unconventional Times 169-198. [[Crossref](#)]
159. Cynthia L. Doniger. 2021. The Ways the Cookie Crumbles: Education and the Margins of Cyclical Adjustment in the Labor Market. *Finance and Economics Discussion Series* **2021**:015, 1-48. [[Crossref](#)]
160. Chunya Bu, John Rogers, Wenbin Wu. 2021. A unified measure of Fed monetary policy shocks. *Journal of Monetary Economics* **118**, 331-349. [[Crossref](#)]
161. Samer Adra. 2021. The conventional and informational impacts of monetary policy on the IPO market. *Economics Letters* **200**, 109751. [[Crossref](#)]
162. UGO ALBERTAZZI, ANDREA NOBILI, FEDERICO M. SIGNORETTI. 2021. The Bank Lending Channel of Conventional and Unconventional Monetary Policy. *Journal of Money, Credit and Banking* **53**:2-3, 261-299. [[Crossref](#)]
163. Michael Smolyansky, Gustavo Suarez. 2021. Monetary Policy and the Corporate Bond Market: How Important is the Fed Information Effect?. *Finance and Economics Discussion Series* **2021**:009, 1-37. [[Crossref](#)]
164. Francisco Luna Santos. 2021. Comparing the impact of discretionary and pre-announced central bank interventions. *Journal of International Money and Finance* **110**, 102307. [[Crossref](#)]
165. Stefan Avdjiev, Bryan Hardy, Patrick McGuire, Goetz Peter. 2021. Home sweet host: A cross-country perspective on prudential and monetary policy spillovers through global banks. *Review of International Economics* **29**:1, 20-36. [[Crossref](#)]

166. Friederike Niepmann, Tim Schmidt-Eisenlohr, Emily Liu. 2021. The effect of US stress tests on monetary policy spillovers to emerging markets. *Review of International Economics* 29:1, 165-194. [[Crossref](#)]
167. Georgia Bush, Tomás Gómez, Alejandro Jara, David Moreno, Konstantin Styryn, Yulia Ushakova. 2021. Macroprudential policy and the inward transmission of monetary policy: The case of Chile, Mexico, and Russia. *Review of International Economics* 29:1, 37-60. [[Crossref](#)]
168. Matthieu Bussière, Jin Cao, Jakob Haan, Robert Hills, Simon Lloyd, Baptiste Meunier, Justine Pedrono, Dennis Reinhardt, Sonalika Sinha, Rhiannon Sowerbutts, Konstantin Styryn. 2021. The interaction between macroprudential policy and monetary policy: Overview. *Review of International Economics* 29:1, 1-19. [[Crossref](#)]
169. Mary Everett, Jakob Haan, David-Jan Jansen, Peter McQuade, Anna Samarina. 2021. Mortgage lending, monetary policy, and prudential measures in small euro-area economies: Evidence from Ireland and the Netherlands. *Review of International Economics* 29:1, 117-143. [[Crossref](#)]
170. Ying Li, Yunpeng Sun, Mengya Chen. 2021. An Evaluation of the Impact of Monetary Easing Policies in Times of a Pandemic. *Frontiers in Public Health* 8. . [[Crossref](#)]
171. Shawn C.-Y. Leu, Mari L. Robertson. 2021. Mortgage credit volumes and monetary policy after the Great Recession. *Economic Modelling* 94, 483-500. [[Crossref](#)]
172. Mahbuba Aktar, Mohammad Zoynul Abedin, Anupam Das Gupta. 2021. The Impact of Monetary Policy Shocks on Corporate Dynamic Investment Activity With Financial Heterogeneity. *SAGE Open* 11:1, 215824402098868. [[Crossref](#)]
173. Cynthia Doniger, Benjamin Kay. 2021. Ten Days Late and Billions of Dollars Short: The Employment Effects of Delays in Paycheck Protection Program Financing. *SSRN Electronic Journal* 53. . [[Crossref](#)]
174. Harrison Shieh. 2021. What We Do In The Shadows: Chinese Shadow Credit Growth and Monetary Policy. *SSRN Electronic Journal* 57. . [[Crossref](#)]
175. Masashige Hamano, Francesco Zanetti. 2021. Monetary Policy, Firm Heterogeneity, and Product Variety. *SSRN Electronic Journal* . [[Crossref](#)]
176. Mattia Bevilacqua, Lukas Brandl-Cheng, Jon Danielsson, Lerby Murat Ergun, Andreas Uthemann, Jean-Pierre Zigrand. 2021. The Calming of Short-Term Market Fears and Its Long-Term Consequences: The Federal Reserve's Reaction to COVID-19. *SSRN Electronic Journal* 66. . [[Crossref](#)]
177. Alina K. Bartscher, Moritz Kuhn, Moritz Schularick, Paul Wachtel. 2021. Monetary Policy and Racial Inequality. *SSRN Electronic Journal* 89. . [[Crossref](#)]
178. Stéphane Lhuissier, Benoît Nguyen. 2021. The Dynamic Effects of the ECB's Asset Purchases: A Survey-Based Identification. *SSRN Electronic Journal* . [[Crossref](#)]
179. Claire Thürwächter. 2021. Heterogeneity in Corporate Debt Structures and the Transmission of Monetary Policy. *SSRN Electronic Journal* 27. . [[Crossref](#)]
180. Michele Piffer. 2021. Monetary policy shocks over the business cycle: Extending the Smooth Transition framework. *SSRN Electronic Journal* 76. . [[Crossref](#)]
181. Martin Bruns, Helmut Lutkepohl. 2021. Comparison of Local Projection Estimators for Proxy Vector Autoregressions. *SSRN Electronic Journal* 36. . [[Crossref](#)]
182. Alina Arefeva, Nikolay Arefyev. 2021. Conventional Monetary Policy Re-Estimated. *SSRN Electronic Journal* 80. . [[Crossref](#)]

183. Federico Di Pace, Christoph Görtz. 2021. Sectoral Comovement, Monetary Policy and the Credit Channel. *SSRN Electronic Journal* 97. . [[Crossref](#)]
184. Fei Shang. 2021. The Effect of Uncertainty on the Sensitivity of the Yield Curve to Monetary Policy Surprises. *SSRN Electronic Journal* 32. . [[Crossref](#)]
185. Lisa-Maria Kampl. 2021. Measuring the Short-Term Effects of the ECB's Unconventional Monetary Policy on Financial Markets: A Review. *Credit and Capital Markets – Kredit und Kapital* 54:1, 37-77. [[Crossref](#)]
186. Isaiah Oino. 2021. Regulatory capital: Implications on credit creation and profitability. *Cogent Economics & Finance* 9:1. . [[Crossref](#)]
187. Ralf Fendel, Frederik Neugebauer. 2020. Country-specific euro area government bond yield reactions to ECB's non-standard monetary policy program announcements. *German Economic Review* 21:4, 417-474. [[Crossref](#)]
188. Refet S. Gürkaynak, Burçin Kısacıkoglu, Jonathan H. Wright. 2020. Missing Events in Event Studies: Identifying the Effects of Partially Measured News Surprises. *American Economic Review* 110:12, 3871-3912. [[Abstract](#)] [[View PDF article](#)] [[PDF with links](#)]
189. Aktham I. Maghyreh, Osama D. Sweidan. 2020. Do structural shocks in the crude oil market affect biofuel prices?. *International Economics* 164, 183-193. [[Crossref](#)]
190. Brent Bundick, A. Lee Smith. 2020. The Dynamic Effects of Forward Guidance Shocks. *The Review of Economics and Statistics* 102:5, 946-965. [[Crossref](#)]
191. Mikhail Mamonov, Vera Pankova, Renat Akhmetov, Anna Pestova. 2020. Financial Shocks and Credit Cycles. *Russian Journal of Money and Finance* 79:4, 45-74. [[Crossref](#)]
192. Jongrim Ha. 2020. Nonlinear transmission of U.S. monetary policy shocks to international financial markets. *International Finance* 23:3, 350-369. [[Crossref](#)]
193. Nicolas Crouzet, Neil R. Mehrotra. 2020. Small and Large Firms over the Business Cycle. *American Economic Review* 110:11, 3549-3601. [[Abstract](#)] [[View PDF article](#)] [[PDF with links](#)]
194. Gene Amromin, Neil Bhutta, Benjamin J. Keys. 2020. Refinancing, Monetary Policy, and the Credit Cycle. *Annual Review of Financial Economics* 12:1, 67-93. [[Crossref](#)]
195. Javier Cuenca, José-Matías Cutillas-Lozano, Domingo Giménez, Alberto Pérez-Bernabeu, José J. López-Espín. 2020. Exploiting Heterogeneous Parallelism on Hybrid Metaheuristics for Vector Autoregression Models. *Electronics* 9:11, 1781. [[Crossref](#)]
196. Christian K. Wolf. 2020. SVAR (Mis)Identification and the Real Effects of Monetary Policy Shocks. *American Economic Journal: Macroeconomics* 12:4, 1-32. [[Abstract](#)] [[View PDF article](#)] [[PDF with links](#)]
197. Bruno Albuquerque, Martin Iseringhausen, Frederic Opitz. 2020. Monetary policy and US housing expansions: The case of time-varying supply elasticities. *Economics Letters* 195, 109471. [[Crossref](#)]
198. Pascal Paul. 2020. The Time-Varying Effect of Monetary Policy on Asset Prices. *The Review of Economics and Statistics* 102:4, 690-704. [[Crossref](#)]
199. Colin Weiss. 2020. Contractionary Devaluation Risk: Evidence from the Free Silver Movement, 1878–1900. *The Review of Economics and Statistics* 102:4, 705-720. [[Crossref](#)]
200. NADAV BEN ZEEV, CHRISTOPHER GUNN, HASHMAT KHAN. 2020. Monetary News Shocks. *Journal of Money, Credit and Banking* 52:7, 1793-1820. [[Crossref](#)]

201. Kurt G. Lunsford. 2020. Policy Language and Information Effects in the Early Days of Federal Reserve Forward Guidance. *American Economic Review* **110**:9, 2899-2934. [[Abstract](#)] [[View PDF article](#)] [[PDF with links](#)]
202. Sebastian K. R  th. 2020. Shifts in monetary policy and exchange rate dynamics: Is Dornbusch's overshooting hypothesis intact, after all?. *Journal of International Economics* **126**, 103344. [[Crossref](#)]
203. Zhigang Ouyang, Fengyu Liu, Ge Zhai, Svitlana Bilan. 2020. Assessment of Resident Happiness under Uncertainty of Economic Policies: Empirical Evidences from China. *Sustainability* **12**:18, 7296. [[Crossref](#)]
204. Fabian Eser, Peter Karadi, Philip R. Lane, Laura Moretti, Chiara Osbat. 2020. The Phillips Curve at the ECB. *The Manchester School* **88**:S1, 50-85. [[Crossref](#)]
205. Jonathan Goldberg, Elizabeth Klee, Edward Simpson Prescott, Paul Wood. 2020. Monetary Policy Strategies and Tools: Financial Stability Considerations. *Finance and Economics Discussion Series* **2020**:074, 1-34. [[Crossref](#)]
206. Stefan Las  en, Andrea Pescatori. 2020. Financial stability and interest-rate policy: A quantitative assessment of costs and benefit. *Canadian Journal of Economics/Revue canadienne d'  conomie* **53**:3, 1246-1273. [[Crossref](#)]
207. Ricardo Lagos, Shengxing Zhang. 2020. Turnover Liquidity and the Transmission of Monetary Policy. *American Economic Review* **110**:6, 1635-1672. [[Abstract](#)] [[View PDF article](#)] [[PDF with links](#)]
208. Adrian Alter, Selim Elekdag. 2020. Emerging market corporate leverage and global financial conditions. *Journal of Corporate Finance* **62**, 101590. [[Crossref](#)]
209. Jiri Slacalek, Oreste Tristani, Giovanni L. Violante. 2020. Household balance sheet channels of monetary policy: A back of the envelope calculation for the euro area. *Journal of Economic Dynamics and Control* **115**, 103879. [[Crossref](#)]
210. Kyungmin Kim, Thomas Laubach, Min Wei. 2020. Macroeconomic Effects of Large-Scale Asset Purchases: New Evidence. *Finance and Economics Discussion Series* **2020**:047. . [[Crossref](#)]
211. Christophe Blot, Paul Hubert, Fabien Labondance. 2020. Monetary policy and asset prices in the euro area since the global financial crisis. *Revue d'  conomie politique* **Vol. 130**:2, 257-281. [[Crossref](#)]
212. Viktorija Cohen, Ar  nas Burinskas. 2020. The Evaluation of the Impact of Macroeconomic Indicators on the Performance of Listed Real Estate Companies and Reits. *Ekonomika* **99**:1, 79-92. [[Crossref](#)]
213. R  diger Bachmann, Sebastian K. R  th. 2020. SYSTEMATIC MONETARY POLICY AND THE MACROECONOMIC EFFECTS OF SHIFTS IN RESIDENTIAL LOAN-TO-VALUE RATIOS. *International Economic Review* **61**:2, 503-530. [[Crossref](#)]
214. Anna A. Pestova, Natalia A. Rostova. 2020. Economic effects of monetary policy in Russia: What do large datasets tell us?. *Voprosy Ekonomiki* :4, 31-53. [[Crossref](#)]
215. Marek Jaroci  ski, Peter Karadi. 2020. Deconstructing Monetary Policy Surprises— The Role of Information Shocks. *American Economic Journal: Macroeconomics* **12**:2, 1-43. [[Abstract](#)] [[View PDF article](#)] [[PDF with links](#)]
216. Ambrogio Cesa-Bianchi, Gregory Thwaites, Alejandro Vicondoa. 2020. Monetary policy transmission in the United Kingdom: A high frequency identification approach. *European Economic Review* **123**, 103375. [[Crossref](#)]

217. Silvia Albrizio, Sangyup Choi, Davide Furceri, Chansik Yoon. 2020. International bank lending channel of monetary policy. *Journal of International Money and Finance* **102**, 102124. [[Crossref](#)]
218. Michiel De Pooter, Giovanni Favara, Michele Modugno, Jason Wu. 2020. Monetary Policy Uncertainty and Monetary Policy Surprises. *Finance and Economics Discussion Series* **2020**:032. . [[Crossref](#)]
219. Hilde C. Bjørnland, Leif Anders Thorsrud, Sepideh Khayati Zahiri. 2020. Do Central Banks Respond Timely to Developments in the Global Economy?. *Oxford Bulletin of Economics and Statistics* **82**:2, 285-310. [[Crossref](#)]
220. Fatih Akçelik, Anıl Talaslı. 2020. Market-based monetary policy expectations for Turkey. *Central Bank Review* **20**:1, 9-19. [[Crossref](#)]
221. Valentina Colombo, Alessia Paccagnini. 2020. Does the credit supply shock have asymmetric effects on macroeconomic variables?. *Economics Letters* **188**, 108958. [[Crossref](#)]
222. Malte Rieth, Jana Wittich. 2020. The impact of ECB policy on structural reforms. *European Economic Review* **122**, 103361. [[Crossref](#)]
223. Adugna Olani. 2020. Dynamic effects of macroeconomic policies on categories of emerging markets' capital inflows. *Macroeconomics and Finance in Emerging Market Economies* **13**:1, 1-28. [[Crossref](#)]
224. Woon Wook Jang. 2020. Risk aversion, uncertainty, and monetary policy: Structural vector autoregressions identified with high-frequency external instruments. *Economics Letters* **186**, 108675. [[Crossref](#)]
225. Francis Leni Anguyo, Rangan Gupta, Kevin Kotzé. 2020. Monetary policy, financial frictions and structural changes in Uganda: a Markov-switching DSGE approach. *Economic Research-Ekonomska Istraživanja* **33**:1, 1538-1561. [[Crossref](#)]
226. Michael McLeay, Silvana Tenreyro. 2020. Optimal Inflation and the Identification of the Phillips Curve. *NBER Macroeconomics Annual* **34**, 199-255. [[Crossref](#)]
227. Francesco Simone Lucidi, Willi Semmler. 2020. Supervisory Shocks to Banks' Credit Standards and Their Macro Impact. *SSRN Electronic Journal* **82**. . [[Crossref](#)]
228. Nataliia Ostapenko. 2020. Identification of Monetary Policy Shocks From Federal Reserve Surprises. *SSRN Electronic Journal* **120**. . [[Crossref](#)]
229. Min Fang. 2020. Lumpy Investment, Uncertainty, and Monetary Policy. *SSRN Electronic Journal* **76**. . [[Crossref](#)]
230. Christoph Boehm, T. Niklas Kroner. 2020. What Does High Frequency Identification Tell Us About the Transmission and Synchronization of Business Cycles?. *SSRN Electronic Journal* . [[Crossref](#)]
231. Nishant Vats. 2020. What Does Financial Heterogeneity Say About the Transmission of Monetary Policy?. *SSRN Electronic Journal* **102**. . [[Crossref](#)]
232. Giulia Rivolta, Carmine Trecroci. 2020. Measuring the Effects of U.S. Uncertainty and Monetary Conditions on EMEs' Macroeconomic Dynamics. *SSRN Electronic Journal* . [[Crossref](#)]
233. Cristiano Cantore, Filippo Ferroni, Miguel Leon-Ledesma. 2020. The Missing Link: Monetary Policy and the Labor Share. *SSRN Electronic Journal* **14**. . [[Crossref](#)]
234. Nicolas Soenen, Rudi Vander Venet. 2020. Drivers of Bank Default Risk: Bank Business Models, the Sovereign and Monetary Policy. *SSRN Electronic Journal* . [[Crossref](#)]
235. Derin Aksit. 2020. How to Model Forward Guidance and Address a Larger Puzzle. *SSRN Electronic Journal* . [[Crossref](#)]

236. Hakan Yilmazkuday. 2020. COVID-19 and Exchange Rates: Spillover Effects of U.S. Monetary Policy. *SSRN Electronic Journal* **134**. . [[Crossref](#)]
237. Nataliia Ostapenko. 2020. Central Bank Communication: Information and Policy Shocks. *SSRN Electronic Journal* **101**. . [[Crossref](#)]
238. Robin Döttling, Lev Ratnovski. 2020. Monetary Policy and Intangible Investment. *SSRN Electronic Journal* **5**. . [[Crossref](#)]
239. Daniel Murphy. 2020. Monetary Policy and Gentrification: the Cost Channel. *SSRN Electronic Journal* **133**. . [[Crossref](#)]
240. Aaron Barkley, David P. Byrne, Xiaosong Wu. 2020. The Price Effects of Shaming Oligopolists: A Study of the COVID-19 Oil Price Shock. *SSRN Electronic Journal* . [[Crossref](#)]
241. Rashad Ahmed. 2020. Global Flight-to-Safety Shocks. *SSRN Electronic Journal* . [[Crossref](#)]
242. Marco Casiraghi, Thomas McGregor, Berardino Palazzo. 2020. Young Firms and Monetary Policy Transmission. *SSRN Electronic Journal* **72**. . [[Crossref](#)]
243. Iftekhar Hasan, Boreum Kwak, Xiang Li. 2020. Financial Technologies and the Effectiveness of Monetary Policy Transmission. *SSRN Electronic Journal* **76**. . [[Crossref](#)]
244. Fernando E. Calles Albiñana, Isabel Martínez Rodríguez. Creating Value Through Entrepreneurship Based on the Economic Cycle 21-42. [[Crossref](#)]
245. María-Soledad Castaño-Martínez. Innovation, Value Creation, and Entrepreneurship by Opportunity 43-63. [[Crossref](#)]
246. . Monetary Policy Operations of Central Banks in the E7 Economies 65-91. [[Crossref](#)]
247. Signe Rosenberg. 2019. The effects of conventional and unconventional monetary policy on house prices in the Scandinavian countries. *Journal of Housing Economics* **46**, 101659. [[Crossref](#)]
248. SIMON GILCHRIST, VIVIAN YUE, EGON ZAKRAJŠEK. 2019. U.S. Monetary Policy and International Bond Markets. *Journal of Money, Credit and Banking* **51**:S1, 127-161. [[Crossref](#)]
249. Christopher S. Armstrong, Stephen Glaeser, John D. Kepler. 2019. Accounting quality and the transmission of monetary policy. *Journal of Accounting and Economics* **68**:2-3, 101265. [[Crossref](#)]
250. Elias Albagli, Luis Ceballos, Sebastian Claro, Damian Romero. 2019. Channels of US monetary policy spillovers to international bond markets. *Journal of Financial Economics* **134**:2, 447-473. [[Crossref](#)]
251. José Rômulo de Castro Vieira, Flavio Barboza, Vinicius Amorim Sobreiro, Herbert Kimura. 2019. Machine learning models for credit analysis improvements: Predicting low-income families' default. *Applied Soft Computing* **83**, 105640. [[Crossref](#)]
252. Simon Gilchrist, Vivian Yue, Egon Zakrajsek. 2019. US Monetary Policy and International Bond Markets. *Finance and Economics Discussion Series* **2019**:014r1. . [[Crossref](#)]
253. Davide Furceri, Fabio Mazzola, Pietro Pizzuto. 2019. Asymmetric effects of monetary policy shocks across US states. *Papers in Regional Science* **98**:5, 1861-1891. [[Crossref](#)]
254. Jing Fang, Dingming Liu. 2019. Foreign official purchases of U.S. treasuries and mortgage rates. *Applied Economics Letters* **26**:16, 1306-1312. [[Crossref](#)]
255. Aeimit Lakdawala. 2019. Decomposing the effects of monetary policy using an external instruments SVAR. *Journal of Applied Econometrics* **34**:6, 934-950. [[Crossref](#)]
256. Giovanni Angelini, Luca Fanelli. 2019. Exogenous uncertainty and the identification of structural vector autoregressions with external instruments. *Journal of Applied Econometrics* **34**:6, 951-971. [[Crossref](#)]

257. Daniel A. Dias, João B. Duarte. 2019. Monetary policy, housing rents, and inflation dynamics. *Journal of Applied Econometrics* **34**:5, 673-687. [[Crossref](#)]
258. Lucia Alessi, Mark Kerssenfischer. 2019. The response of asset prices to monetary policy shocks: Stronger than thought. *Journal of Applied Econometrics* **34**:5, 661-672. [[Crossref](#)]
259. SAMI ALPANDA, SARAH ZUBAIRY. 2019. Household Debt Overhang and Transmission of Monetary Policy. *Journal of Money, Credit and Banking* **51**:5, 1265-1307. [[Crossref](#)]
260. MATHIAS KLEIN, LUDGER LINNEMANN. 2019. Macroeconomic Effects of Government Spending: The Great Recession was (Really) Different. *Journal of Money, Credit and Banking* **51**:5, 1237-1264. [[Crossref](#)]
261. Carsten Jentsch, Kurt G. Lunsford. 2019. The Dynamic Effects of Personal and Corporate Income Tax Changes in the United States: Comment. *American Economic Review* **109**:7, 2655-2678. [[Abstract](#)] [[View PDF article](#)] [[PDF with links](#)]
262. Karel Mertens, Morten O. Ravn. 2019. The Dynamic Effects of Personal and Corporate Income Tax Changes in the United States: Reply. *American Economic Review* **109**:7, 2679-2691. [[Abstract](#)] [[View PDF article](#)] [[PDF with links](#)]
263. Peter Karadi, Adam Reiff. 2019. Menu Costs, Aggregate Fluctuations, and Large Shocks. *American Economic Journal: Macroeconomics* **11**:3, 111-146. [[Abstract](#)] [[View PDF article](#)] [[PDF with links](#)]
264. Björn Richter, Moritz Schularick, Ilhyock Shim. 2019. The costs of macroprudential policy. *Journal of International Economics* **118**, 263-282. [[Crossref](#)]
265. Bora Durdu, Alex Martin, Ilknur Zer. 2019. The Role of U.S. Monetary Policy in Global Banking Crises. *Finance and Economics Discussion Series* **2019**:039. . [[Crossref](#)]
266. BERNARDO MORAIS, JOSÉ-LUIS PEYDRÓ, JESSICA ROLDÁN-PENÁ, CLAUDIA RUIZ-ORTEGA. 2019. The International Bank Lending Channel of Monetary Policy Rates and QE: Credit Supply, Reach-for-Yield, and Real Effects. *The Journal of Finance* **74**:1, 55-90. [[Crossref](#)]
267. . Bibliographie 261-270. [[Crossref](#)]
268. Dario Caldara, Edward Herbst. 2019. Monetary Policy, Real Activity, and Credit Spreads: Evidence from Bayesian Proxy SVARs. *American Economic Journal: Macroeconomics* **11**:1, 157-192. [[Abstract](#)] [[View PDF article](#)] [[PDF with links](#)]
269. Ki Young Park, Youngjoon Lee, Soohyon Kim. 2019. Deciphering Monetary Policy Board Minutes Through Text Mining Approach: The Case of Korea. *SSRN Electronic Journal* . [[Crossref](#)]
270. Kwangyong Park. 2019. Uncertainty, Attention Allocation and Monetary Policy Asymmetry. *SSRN Electronic Journal* . [[Crossref](#)]
271. Hengjie Ai, Leyla Jianyu Han, Xuhui (Nick) Pan, Lai Xu. 2019. The Cross-Section of Monetary Policy Announcement Premium. *SSRN Electronic Journal* **48**. . [[Crossref](#)]
272. Gee Hee Hong, Matthew Klepacz, Ernesto Pastén, Raphael Schoenle. 2019. From Micro to Macro: A New Methodology to Discriminate Among Models. *SSRN Electronic Journal* . [[Crossref](#)]
273. Young Joon Lee, Soohyon Kim, Ki Young Park. 2019. Measuring Monetary Policy Surprises Using Text Mining: The Case of Korea. *SSRN Electronic Journal* . [[Crossref](#)]
274. Seohyun Lee, Rickard Nyman. 2019. Tracking Uncertainty Through the Relative Sentiment Shift Series. *SSRN Electronic Journal* . [[Crossref](#)]

275. Ren Zhang. 2019. News Shocks and the Effects of Monetary Policy. *SSRN Electronic Journal* 108. . [[Crossref](#)]
276. Oliver Boguth, Vincent Gregoire, Charles Martineau. 2019. Price Pressure and Efficiency on FOMC Announcements. *SSRN Electronic Journal* . [[Crossref](#)]
277. Anna Samarina, Anh D. M. Nguyen. 2019. Does Monetary Policy Affect Income Inequality in the Euro Area?. *SSRN Electronic Journal* . [[Crossref](#)]
278. Subhendu Bhowal. 2019. Heterogenous Banks and Macroprudential Regulations. *SSRN Electronic Journal* . [[Crossref](#)]
279. Donghai Zhang. 2019. Term Structure, Forecast Revision and the Signaling Channel of Monetary Policy. *SSRN Electronic Journal* 54. . [[Crossref](#)]
280. James Cloyne, Clodomiro Ferreira, Maren Froemel, Paolo Surico. 2019. Monetary Policy, Corporate Finance and Investment. *SSRN Electronic Journal* . [[Crossref](#)]
281. Marco Giacomelli, Rodney Ramcharan, Edison Yu. 2019. The FOMC and the Mortgage Market. *SSRN Electronic Journal* . [[Crossref](#)]
282. Zilong Niu. 2019. Underreaction to Macroeconomic News Announcements and the Downward-Sloping Security Market Line. *SSRN Electronic Journal* . [[Crossref](#)]
283. Ricardo J. Caballero, Alp Simsek. 2019. Prudential Monetary Policy. *SSRN Electronic Journal* . [[Crossref](#)]
284. Niels D. Gilbert. 2019. Euro Area Sovereign Risk Spillovers Before and after the Ecb's OMT Announcement. *SSRN Electronic Journal* . [[Crossref](#)]
285. Daniel Lewis. 2019. Announcement-Specific Decompositions of Unconventional Monetary Policy Shocks and Their Macroeconomic Effects. *SSRN Electronic Journal* . [[Crossref](#)]
286. Robert Jarrow, Sujan Lamichhane. 2019. Risk Premia, Asset Price Bubbles, and Monetary Policy. *SSRN Electronic Journal* 29. . [[Crossref](#)]
287. Olivier Darmouni, Oliver Giesecke, Alexander Rodnyansky. 2019. The Bond Lending Channel of Monetary Policy. *SSRN Electronic Journal* . [[Crossref](#)]
288. Karthik Sastry. 2019. Public Sentiments and Monetary Surprises. *SSRN Electronic Journal* . [[Crossref](#)]
289. Francesco Simone Lucidi, Willi Semmler. 2019. Nonlinear Credit Dynamics, Regime Switches in the Output Gap and Supervisory Shocks. *SSRN Electronic Journal* 4. . [[Crossref](#)]
290. Daniel J. Lewis, Christos Makridis, Karel Mertens. 2019. Do Monetary Policy Announcements Shift Household Expectations?. *SSRN Electronic Journal* 106. . [[Crossref](#)]
291. Carlo Altavilla, Lorenzo Burlon, Mariassunta Giannetti, Sarah Holton. 2019. Is There a Zero Lower Bound? The Effects of Negative Policy Rates on Banks and Firms. *SSRN Electronic Journal* 31. . [[Crossref](#)]
292. Zhengyang Chen. 2019. The Long-term Rate and Interest Rate Volatility in Monetary Policy Transmission. *SSRN Electronic Journal* . [[Crossref](#)]
293. Ashish Garg, Christian L. Goulding, Campbell R. Harvey, Michele Mazzoleni. 2019. Momentum Turning Points. *SSRN Electronic Journal* . [[Crossref](#)]
294. Jonathan R. Stroud, Hang Zhou. 2019. Monetary Policy in a Fixed Exchange Rate Regime: Structural Vector Autoregressions with Jumps. *SSRN Electronic Journal* . [[Crossref](#)]
295. Wenting Liao, Jun Ma, Chengsi Zhang. 2019. Exchange Rate Dynamics and Global Monetary Policy Spillovers: Time-Varying Dynamic Causal Effects. *SSRN Electronic Journal* . [[Crossref](#)]

296. Mark Kerssenfischer. 2019. The puzzling effects of monetary policy in VARs: Invalid identification or missing information?. *Journal of Applied Econometrics* **34**:1, 18-25. [[Crossref](#)]
297. José Luis Montiel Olea, Mikkel Plagborg-Møller. 2019. Simultaneous confidence bands: Theory, implementation, and an application to SVARs. *Journal of Applied Econometrics* **34**:1, 1-17. [[Crossref](#)]
298. Martin Feldkircher, Florian Huber. 2018. Unconventional U.S. Monetary Policy: New Tools, Same Channels?. *Journal of Risk and Financial Management* **11**:4, 71. [[Crossref](#)]
299. JOHN H. ROGERS, CHIARA SCOTTI, JONATHAN H. WRIGHT. 2018. Unconventional Monetary Policy and International Risk Premia. *Journal of Money, Credit and Banking* **50**:8, 1827-1850. [[Crossref](#)]
300. Vincent Sterk, Silvana Tenreyro. 2018. The transmission of monetary policy through redistributions and durable purchases. *Journal of Monetary Economics* **99**, 124-137. [[Crossref](#)]
301. Itamar Drechsler, Alexi Savov, Philipp Schnabl. 2018. Liquidity, Risk Premia, and the Financial Transmission of Monetary Policy. *Annual Review of Financial Economics* **10**:1, 309-328. [[Crossref](#)]
302. Juan Antolín-Díaz, Juan F. Rubio-Ramírez. 2018. Narrative Sign Restrictions for SVARs. *American Economic Review* **108**:10, 2802-2829. [[Abstract](#)] [[View PDF article](#)] [[PDF with links](#)]
303. Nicholas Apergis, James E. Payne. 2018. Monetary policy rules and the equity risk premium: Evidence from the US experience. *Review of Financial Economics* **36**:4, 287-299. [[Crossref](#)]
304. Yannick Timmer. 2018. Emerging market corporate bond yields and monetary policy. *Emerging Markets Review* **36**, 130-143. [[Crossref](#)]
305. Lawrence J. Christiano, Martin S. Eichenbaum, Mathias Trabandt. 2018. On DSGE Models. *Journal of Economic Perspectives* **32**:3, 113-140. [[Abstract](#)] [[View PDF article](#)] [[PDF with links](#)]
306. Emi Nakamura, Jón Steinsson. 2018. Identification in Macroeconomics. *Journal of Economic Perspectives* **32**:3, 59-86. [[Abstract](#)] [[View PDF article](#)] [[PDF with links](#)]
307. Laurent Ferrara, Pierre Guérin. 2018. What are the macroeconomic effects of high-frequency uncertainty shocks?. *Journal of Applied Econometrics* **33**:5, 662-679. [[Crossref](#)]
308. JEREMY C. STEIN, ADI SUNDERAM. 2018. The Fed, the Bond Market, and Gradualism in Monetary Policy. *The Journal of Finance* **73**:3, 1015-1060. [[Crossref](#)]
309. Michiel De Pooter, Giovanni Favara, Michele Modugno, Jason Wu. 2018. Monetary Policy Surprises and Monetary Policy Uncertainty. *FEDS Notes* **2018**:2176. . [[Crossref](#)]
310. Michael Hachula, Dieter Nautz. 2018. The dynamic impact of macroeconomic news on long-term inflation expectations. *Economics Letters* **165**, 39-43. [[Crossref](#)]
311. Greg Kaplan, Benjamin Moll, Giovanni L. Violante. 2018. Monetary Policy According to HANK. *American Economic Review* **108**:3, 697-743. [[Abstract](#)] [[View PDF article](#)] [[PDF with links](#)]
312. Benjamin Nelson, Gabor Pinter, Konstantinos Theodoridis. 2018. Do contractionary monetary policy shocks expand shadow banking?. *Journal of Applied Econometrics* **33**:2, 198-211. [[Crossref](#)]
313. A. A. Pestova. 2018. On the effects of monetary policy in Russia: The role of the space of spanned shocks and the policy regime shifts. *Voprosy Ekonomiki* :2, 33-55. [[Crossref](#)]
314. Simon Gilchrist, Vivian Yue, Egon Zakrajsek. 2018. US Monetary Policy and International Bond Markets. *Finance and Economics Discussion Series* **2018**:014. . [[Crossref](#)]
315. ITAMAR DRECHSLER, ALEXI SAVOV, PHILIPP SCHNABL. 2018. A Model of Monetary Policy and Risk Premia. *The Journal of Finance* **73**:1, 317-373. [[Crossref](#)]
316. Benjamin Garcia, Arsenios Skaperdas. 2018. Inferring the Shadow Rate from Real Activity. *SSRN Electronic Journal* **1**. . [[Crossref](#)]

317. Priit Jeenas. 2018. Monetary Policy Shocks, Financial Structure, and Firm Activity: A Panel Approach. *SSRN Electronic Journal* . [[Crossref](#)]
318. Manfred M. Fischer, Florian Huber, Michael Pfarrhofer, Petra Staufer. 2018. The Dynamic Impact of Monetary Policy on Regional Housing Prices in the US: Evidence Based on Factor-augmented Vector Autoregressions. *SSRN Electronic Journal* . [[Crossref](#)]
319. Tony Zhang. 2018. Monetary Policy Spillovers Through Invoicing Currencies. *SSRN Electronic Journal* . [[Crossref](#)]
320. Simon Lloyd. 2018. Overnight Index Swap Market-Based Measures of Monetary Policy Expectations. *SSRN Electronic Journal* . [[Crossref](#)]
321. Felix Geiger, Fabian Schupp. 2018. With a Little Help from My Friends: Survey-Based Derivation of Euro Area Short Rate Expectations at the Effective Lower Bound. *SSRN Electronic Journal* . [[Crossref](#)]
322. Martina Cecioni. 2018. ECB Monetary Policy and the Euro Exchange Rate. *SSRN Electronic Journal* . [[Crossref](#)]
323. Giovanni Angelini, Luca Fanelli. 2018. Identification and Estimation Issues in Structural Vector Autoregressions with External Instruments. *SSRN Electronic Journal* . [[Crossref](#)]
324. Haifeng Guo, Chi-Hsiou Daniel Hung, Alexandros Kontonikas. 2018. The Fed and the Stock Market: A Tale of Sentiment States. *SSRN Electronic Journal* **120**. . [[Crossref](#)]
325. Diego R. KKnzig. 2018. The Macroeconomic Effects of Oil Supply Shocks: New Evidence from OPEC Announcements. *SSRN Electronic Journal* . [[Crossref](#)]
326. Zehao Li. 2018. Leverage of the Intermediary and the Transmission of Monetary Policy. *SSRN Electronic Journal* . [[Crossref](#)]
327. Gabor Pinter. 2018. Macroeconomic Shocks and Risk Premia. *SSRN Electronic Journal* **104**. . [[Crossref](#)]
328. Ursula Wiriadinata. 2018. External Debt, Currency Risk, and International Monetary Policy Transmission. *SSRN Electronic Journal* . [[Crossref](#)]
329. Xu Zhang. 2018. Evaluating the Effects of Forward Guidance and Large-Scale Asset Purchases. *SSRN Electronic Journal* . [[Crossref](#)]
330. Andreas Tryphonides. 2018. Inflation Determination and Identification: Some Positive Results. *SSRN Electronic Journal* **42**. . [[Crossref](#)]
331. Björn Richter, Moritz Schularick, Ilhyock Shim. 2018. The Macroeconomic Effects of Macprudential Policy. *SSRN Electronic Journal* **33**. . [[Crossref](#)]
332. Youngjoon Lee, Soohyon Kim, Ki Young Park. 2018. Deciphering Monetary Policy Committee Minutes with Text Mining Approach: A Case of South Korea. *SSRN Electronic Journal* . [[Crossref](#)]
333. Mathias Klein, Ludger Linnemann. 2018. Macroeconomic Effects of Government Spending: The Great Recession Was (Really) Different. *SSRN Electronic Journal* **59**. . [[Crossref](#)]
334. Seohyun Lee, Inhwan So, Jongrim Ha. 2018. Identifying Uncertainty Shocks Due to Geopolitical Swings in Korea. *SSRN Electronic Journal* . [[Crossref](#)]
335. Vaishali Garga, Sanjay Singh. 2018. Output Hysteresis and Optimal Monetary Policy. *SSRN Electronic Journal* **10**. . [[Crossref](#)]
336. Regis Barnichon, Geert Mesters. 2018. Identifying the Phillips Curve from Shifts in Demand. *SSRN Electronic Journal* . [[Crossref](#)]

337. Michael Ellington. 2018. The Empirical Relevance of the Shadow Rate and the Zero Lower Bound. *SSRN Electronic Journal* . [[Crossref](#)]
338. Ethan Min Fang. 2018. Monetary Policy Transmission Through Financial Heterogeneity: Firm-Level Evidence. *SSRN Electronic Journal* **127**. . [[Crossref](#)]
339. Maria Sandström. 2018. The Impact of Monetary Policy on Household Borrowing - a High-Frequency IV Identification. *SSRN Electronic Journal* **43**. . [[Crossref](#)]
340. Helmut Lütkepohl, Anna Staszewska-Bystrova, Peter Winker. 2018. Constructing Joint Confidence Bands for Impulse Response Functions of VAR Models: A Review. *SSRN Electronic Journal* . [[Crossref](#)]
341. Kurt G. Lunsford. 2018. Understanding the Aspects of Federal Reserve Forward Guidance. *SSRN Electronic Journal* . [[Crossref](#)]
342. Helmut Herwartz, Hannes Rohloff. 2018. Less Bang for the Buck? Assessing the Role of Inflation Uncertainty for U.S. Monetary Policy Transmission in a Data Rich Environment. *SSRN Electronic Journal* . [[Crossref](#)]
343. Xu Zhang. 2018. A New Measure of Monetary Policy Shocks. *SSRN Electronic Journal* **2012**. . [[Crossref](#)]
344. Rong Fu. 2018. Financial Uncertainty and the Effectiveness of Monetary Policy. *SSRN Electronic Journal* . [[Crossref](#)]
345. Raymond Kim. 2018. Better Than Risk-Free: Do Reserve Premiums Crowd out Bank Lending?. *SSRN Electronic Journal* **20**. . [[Crossref](#)]
346. Kyungmin Kim. 2017. Identification of Monetary Policy Shocks with External Instrument SVAR. *Finance and Economics Discussion Series* **2017**:113. . [[Crossref](#)]
347. G. Andrew Karolyi, Kirsty J. McLaren. 2017. Racing to the exits: International transmissions of funding shocks during the Federal Reserve's taper experiment. *Emerging Markets Review* **32**, 96-115. [[Crossref](#)]
348. Sebastian K. Rüth. 2017. State-dependent monetary policy transmission and financial market tensions. *Economics Letters* **157**, 56-61. [[Crossref](#)]
349. James H. Stock, Mark W. Watson. 2017. Twenty Years of Time Series Econometrics in Ten Pictures. *Journal of Economic Perspectives* **31**:2, 59-86. [[Abstract](#)] [[View PDF article](#)] [[PDF with links](#)]
350. Luca Dedola, Giulia Rivolta, Livio Stracca. 2017. If the Fed sneezes, who catches a cold?. *Journal of International Economics* **108**, S23-S41. [[Crossref](#)]
351. . Regional Outlooks 81-190. [[Crossref](#)]
352. Tatjana Dahlhaus. 2017. Conventional Monetary Policy Transmission During Financial Crises: An Empirical Analysis. *Journal of Applied Econometrics* **32**:2, 401-421. [[Crossref](#)]
353. Mariusz Kapuściński. 2017. The Role of Bank Balance Sheets in Monetary Policy Transmission: Evidence from Poland. *Eastern European Economics* **55**:1, 50-69. [[Crossref](#)]
354. Jeffrey R. Campbell, Jonas D. M. Fisher, Alejandro Justiniano, Leonardo Melosi. 2017. Forward Guidance and Macroeconomic Outcomes since the Financial Crisis. *NBER Macroeconomics Annual* **31**:1, 283-357. [[Crossref](#)]
355. Giovanni Pellegrino. 2017. Uncertainty and Monetary Policy in the US: A Journey into Non-Linear Territory. *SSRN Electronic Journal* **76**. . [[Crossref](#)]
356. Paul Hubert, Labondance Fabien. 2017. Central Bank Sentiment and Policy Expectations. *SSRN Electronic Journal* **81**. . [[Crossref](#)]

357. Hilde C. Bjørnland, Leif Anders Thorsrud, Sepideh Khayati Zahiri. 2017. Do Central Banks Respond Timely to Developments in the Global Economy?. *SSRN Electronic Journal* 118. . [\[Crossref\]](#)
358. Daniel Andrei, Bernard Herskovic, Olivier Ledoit. 2017. The Redistributive Effects of Monetary Policy. *SSRN Electronic Journal* . [\[Crossref\]](#)
359. Giovanni Caggiano, Efrem Castelnuovo. 2017. Uncertainty and Monetary Policy in Good and Bad Times. *SSRN Electronic Journal* . [\[Crossref\]](#)
360. Aeimit Lakdawala, Matthew Schaffer. 2017. Federal Reserve Private Information and the Stock Market. *SSRN Electronic Journal* . [\[Crossref\]](#)
361. Silvia Miranda-Agrippino, Giovanni Ricco. 2017. The Transmission of Monetary Policy Shocks. *SSRN Electronic Journal* 53. . [\[Crossref\]](#)
362. Kiyotaka Nakashima, Masahiko Shibamoto, Koji Takahashi. 2017. Identifying Unconventional Monetary Policy Shocks. *SSRN Electronic Journal* 81. . [\[Crossref\]](#)
363. Falk Brruning, Victoria Ivashina. 2017. U.S. Monetary Policy and Emerging Markets Credit Cycles. *SSRN Electronic Journal* 1. . [\[Crossref\]](#)
364. Fernando Ballabriga, Karen Davtyan. 2017. The Distributive Impact of Monetary Policy: Evidence from the UK. *SSRN Electronic Journal* 54. . [\[Crossref\]](#)
365. Adam Golinski. 2017. Monetary Policy at the Zero Lower Bound: Information in the Federal Reserve's Balance Sheet. *SSRN Electronic Journal* . [\[Crossref\]](#)
366. Assia Elgouacem, Riccardo Zago. 2017. Share Buybacks, Monetary Policy and the Cost of Debt. *SSRN Electronic Journal* . [\[Crossref\]](#)
367. Kyungmin Kim. 2017. Identification of Forward Guidance Shocks with External Instrument SVAR. *SSRN Electronic Journal* . [\[Crossref\]](#)
368. Kiyotaka Nakashima, Masahiko Shibamoto, Koji Takahashi. 2017. Risk-Taking Channel of Unconventional Monetary Policies in Bank Lending. *SSRN Electronic Journal* . [\[Crossref\]](#)
369. Ye Li. 2017. Procyclical Finance: The Money View. *SSRN Electronic Journal* 567. . [\[Crossref\]](#)
370. Ambrogio Cesa-Bianchi, Andrej Sokol. 2017. Financial Shocks, Credit Spreads and the International Credit Channel. *SSRN Electronic Journal* 48. . [\[Crossref\]](#)
371. Tim Alexander Kroencke, Maik Schmeling, Andreas Schrimpf. 2017. The FOMC Risk Shift. *SSRN Electronic Journal* . [\[Crossref\]](#)
372. Inhwan So. 2017. Bank Globalization and Monetary Policy Transmission in Small Open Economies. *SSRN Electronic Journal* 26. . [\[Crossref\]](#)
373. Thorsten Franz. 2017. Monetary Policy, Housing, and Collateral Constraints. *SSRN Electronic Journal* 127. . [\[Crossref\]](#)
374. Martin Floddn, Matilda Kilström, Jjsef Sigurdsson, Roine Vestman. 2017. Household Debt and Monetary Policy: Revealing the Cash-Flow Channel. *SSRN Electronic Journal* 115. . [\[Crossref\]](#)
375. Max Hanisch. 2017. US Monetary Policy and the Euro Area. *SSRN Electronic Journal* . [\[Crossref\]](#)
376. Pietro Cova, Filippo Natoli. 2017. International Financial Flows and the Risk-Taking Channel. *SSRN Electronic Journal* . [\[Crossref\]](#)
377. Joao F. Gomes, Marco Grotteria, Jessica A. Wachter. 2017. Foreseen Risks. *SSRN Electronic Journal* . [\[Crossref\]](#)
378. Nicolas Andreoulis. 2017. The Aggregate Effects of Fiscal Policy in the United Kingdom. *SSRN Electronic Journal* . [\[Crossref\]](#)

379. J. David López-Salido, Jeremy C. Stein, Egon Zakrajsek. 2017. Credit-Market Sentiment and the Business Cycle. *Finance and Economics Discussion Series* 2015:028r1. . [[Crossref](#)]
380. Vasco Cúrdia, Michael Woodford. 2016. Credit Frictions and Optimal Monetary Policy. *Journal of Monetary Economics* 84, 30-65. [[Crossref](#)]
381. James Cloyne, Patrick Hürtgen. 2016. The Macroeconomic Effects of Monetary Policy: A New Measure for the United Kingdom. *American Economic Journal: Macroeconomics* 8:4, 75-102. [[Abstract](#)] [[View PDF article](#)] [[PDF with links](#)]
382. Neil Bhutta, Benjamin J. Keys. 2016. Interest Rates and Equity Extraction During the Housing Boom. *American Economic Review* 106:7, 1742-1774. [[Abstract](#)] [[View PDF article](#)] [[PDF with links](#)]
383. Masazumi Hattori, Andreas Schrimpf, Vladyslav Sushko. 2016. The Response of Tail Risk Perceptions to Unconventional Monetary Policy. *American Economic Journal: Macroeconomics* 8:2, 111-136. [[Abstract](#)] [[View PDF article](#)] [[PDF with links](#)]
384. V.A. Ramey. Macroeconomic Shocks and Their Propagation 71-162. [[Crossref](#)]
385. J.H. Stock, M.W. Watson. Dynamic Factor Models, Factor-Augmented Vector Autoregressions, and Structural Vector Autoregressions in Macroeconomics 415-525. [[Crossref](#)]
386. Brent Bundick, Andrew Smith. 2016. The Dynamic Effects of Forward Guidance Shocks. *SSRN Electronic Journal* 60. . [[Crossref](#)]
387. Nuno Palma. 2016. The Existence and Persistence of Liquidity Effects: Evidence from a Large-Scale Historical Natural Experiment. *SSRN Electronic Journal* 93. . [[Crossref](#)]
388. Martin Flodden, Matilda Kilström, Josef Sigurdsson, Roine Vestman. 2016. Household Debt and Monetary Policy: Revealing the Cash-Flow Channel. *SSRN Electronic Journal* . [[Crossref](#)]
389. Fructuoso Borrillo, Javier Vallés. 2016. The Effects of US Unconventional Monetary Policies in Latin America. *SSRN Electronic Journal* . [[Crossref](#)]
390. Ricardo Lagos, Shengxing Zhang. 2016. Turnover Liquidity and the Transmission of Monetary Policy. *SSRN Electronic Journal* . [[Crossref](#)]
391. Christoph Bertsch, Isaiah Hull, Xin Zhang. 2016. Fed Liftoff and Subprime Loan Interest Rates: Evidence From the Peer-to-Peer Lending Market. *SSRN Electronic Journal* . [[Crossref](#)]
392. Piergiorgio Alessandri, Antonio Maria Conti, Fabrizio Venditti. 2016. The Financial Stability Dark Side of Monetary Policy. *SSRN Electronic Journal* . [[Crossref](#)]
393. Sandra Eickmeier, Norbert Metiu, Esteban Prieto. 2016. Time-Varying Volatility, Financial Intermediation and Monetary Policy. *SSRN Electronic Journal* 69. . [[Crossref](#)]
394. George Andrew Karolyi, Kirsty McLaren. 2016. Racing to the Exits: International Transmissions of Funding Shocks During the Federal Reserve's Taper Experiment. *SSRN Electronic Journal* . [[Crossref](#)]
395. Michael Hachula, Michele Piffer, Malte Rieth. 2016. Unconventional Monetary Policy, Fiscal Side Effects and Euro Area (Im)balances. *SSRN Electronic Journal* . [[Crossref](#)]
396. Jongrim Ha. 2016. International transmission of U.S. monetary policy shocks in open financial markets. *SSRN Electronic Journal* . [[Crossref](#)]
397. Kyunghun Kim. 2016. International Transmission of U.S. Monetary Policy Surprises. *SSRN Electronic Journal* 83. . [[Crossref](#)]
398. Lenno Uuskkla. 2016. Monetary Transmission Mechanism with Firm Turnover. *SSRN Electronic Journal* 14. . [[Crossref](#)]

399. Angela Abbate, Esteban Prieto. 2016. Financial Shocks and Inflation Dynamics. *SSRN Electronic Journal* **625**. . [[Crossref](#)]
400. Ambrogio Cesa-Bianchi, Gregory Thwaites. 2016. Monetary Policy Transmission in an Open Economy: New Data and Evidence from the United Kingdom. *SSRN Electronic Journal* **43**. . [[Crossref](#)]
401. Benjamin Beckers, Kerstin Bernoth. 2016. Monetary Policy and Mispricing in Stock Markets. *SSRN Electronic Journal* . [[Crossref](#)]
402. Mariusz Kapuściński. 2016. The Role of Bank Balance Sheets in Monetary Policy Transmission. Evidence from Poland. *SSRN Electronic Journal* . [[Crossref](#)]
403. Giovanni Caggiano, Efrem Castelnuovo, Antoine Parent, Giovanni Pellegrino. 2016. Liquidity Traps and Large-Scale Financial Crises. *SSRN Electronic Journal* **76**. . [[Crossref](#)]
404. Marco Bassetto, Luca Benzoni, Trevor Serrao. 2016. The Interplay between Financial Conditions and Monetary Policy Shocks. *SSRN Electronic Journal* . [[Crossref](#)]
405. Andreas Neuhierl, Michael Weber. 2016. Monetary Policy and the Stock Market: Time-Series Evidence. *SSRN Electronic Journal* . [[Crossref](#)]
406. Matteo Leombroni, Gyuri Venter, Paul Whelan. 2016. Central Bank Communication and the Yield Curve. *SSRN Electronic Journal* . [[Crossref](#)]
407. Gabor Pinter. 2016. The Macroeconomic Shock with the Highest Price of Risk. *SSRN Electronic Journal* **94**. . [[Crossref](#)]
408. Wongi Kim. 2016. Government Spending Policy Uncertainty and Economic Activity: U.S. Time Series Evidence. *SSRN Electronic Journal* . [[Crossref](#)]
409. Hilde C. Bjørnland, Leif Anders Thorsrud, Sepideh Khayati Zahiri. 2016. Do Central Banks Respond Timely to Developments in the Global Economy?. *SSRN Electronic Journal* **118**. . [[Crossref](#)]
410. Aemitt Lakdawala. 2016. Decomposing the Effects of Monetary Policy Using an External Instruments SVAR. *SSRN Electronic Journal* . [[Crossref](#)]
411. Andrew L. Detzel. 2015. Monetary Policy Surprises, Investment Opportunities, and Asset Prices. *SSRN Electronic Journal* . [[Crossref](#)]
412. David Lopez-Salido, Jeremy C. Stein, Egon Zakrajsek. 2015. Credit-Market Sentiment and the Business Cycle. *SSRN Electronic Journal* . [[Crossref](#)]
413. Martin Beraja, Andreas Fuster, Erik Hurst, Joseph Vavra. 2015. Regional Heterogeneity and Monetary Policy. *SSRN Electronic Journal* . [[Crossref](#)]
414. Maik Schmeling, Christian Wagner. 2015. Does Central Bank Tone Move Asset Prices?. *SSRN Electronic Journal* . [[Crossref](#)]
415. Salem M. Abo-Zaid. 2015. The Government Spending Multiplier in a Model with the Cost Channel. *SSRN Electronic Journal* . [[Crossref](#)]
416. Nina Boyarchenko, Valentin Haddad, Matthew C. Plosser. 2015. Market Confidence and Monetary Policy. *SSRN Electronic Journal* **8**. . [[Crossref](#)]
417. Michele Piffer. 2015. Monetary Policy and Defaults in the US. *SSRN Electronic Journal* **52**. . [[Crossref](#)]